Quantifying the Sensitivity of EU Debt Sustainability Analysis

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Abstract

The reformed EU fiscal rules derive each country's required fiscal adjustment from a debt sustainability analysis (DSA). Using a replication of the Commission's 2024 reference trajectories, this paper quantifies how the resulting fiscal targets depend on the inputs of the DSA. Because the target is the highest balance required by several criteria, an input matters only where it affects the binding criterion. Drawing forecasts and parameters jointly from past forecast errors and published estimates, I find that the target rises by 0.65 percentage points of GDP at the 75th percentile and falls by 0.36 points at the 25th, compared with an average required adjustment of 2.6 points over four years. With crisis-level multipliers well above one, the debt sustainability safeguard can require implausibly large adjustments or become impossible to meet. By contrast, the stochastic criterion rarely binds, so its specification matters little. Finally, modelling choices reflecting the degree of tolerable risk can move the targets by about as much as uncertainty about forecasts and parameters.

JEL classification: E62, H62, H63, H68

Keywords: debt sustainability analysis, EU fiscal rules, sensitivity analysis, forecast errors, fiscal multiplier

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1 Introduction

Since April 2024, the EU fiscal rules have set fiscal targets for each member state through a debt sustainability analysis (DSA). Member states with government debt above 60 percent of GDP or a deficit above 3 percent of GDP receive from the European Commission a reference trajectory for net expenditure over an adjustment period of four to seven years. The trajectory leads to a target for the structural primary balance (SPB), the primary balance corrected for the cycle and one-off measures, at the end of the adjustment period. The Commission sets the SPB target so that debt declines, or stays below 60 percent of GDP, in a baseline projection, in stress tests and in a stochastic simulation. Numerical safeguards and the excessive deficit procedure (EDP) can raise the target further.

The DSA makes these targets country-specific, but it also makes them depend on multiple forecasts, parameters and methodological choices. Governments can argue for different inputs in their medium-term fiscal plans, at times with large consequences for the required adjustment (Zettelmeyer et al., 2025), and the regulation foresees a review of the DSA methodology (European Parliament and Council of the European Union, 2024; Boivin and Darvas, 2025). Both the assessment of national plans and the review require an understanding of how much the targets depend on each input.

This paper quantifies the sensitivity of the SPB targets of the 27 EU member states to the inputs of the Commission's DSA. It separates three questions. The first is how much the targets depend on economic forecasts and model parameters. The second is how much they depend on the method of the stochastic analysis. The third is how much they depend on the calibration of risk tolerance, that is, on how much risk the rules accept. It uses the replication of the DSA that we present in Darvas et al. (2023, 2024). This replication reproduces the reference trajectories of the Commission's prior guidance of 2024. I divide the inputs into four groups: forecasts, model parameters, the specification of the stochastic analysis, and the calibration of risk tolerance. I vary inputs both individually and jointly, by taking draws from the distributions of forecast errors and published parameters. For each change, I report the effect on the target required by the DSA criteria, as well as on the binding target, which in addition meets the safeguards and the minimum adjustment of the EDP. Because the target is the highest of several requirements, an input moves it only through the binding criterion. In the baseline, the deficit criterion and the adverse interest-growth stress test determine the DSA target for 23 of the 27 countries. The effects of the inputs are therefore concentrated on these two criteria. Because a favourable input lowers the target only until another criterion binds, whereas an adverse input raises it without such a limit, inputs affect targets asymmetrically.

The first group of model inputs contains the forecasts that determine the central projection of debt: interest rates and inflation ten years ahead, potential growth, and ageing costs. The second group contains the model parameters that link the SPB to output: the fiscal multiplier, the persistence of the multiplier effect, and the budget semi-elasticity. I first shift each forecast

by the size of a typical past forecast error and vary each parameter within the range of published estimates. Potential growth and the long-term interest rate ten years ahead have the largest effects. I then draw the forecasts and the parameters jointly from the distribution of past forecast errors and published parameter estimates. I define favourable and adverse combinations of assumptions as the 25th and 75th percentiles of the resulting targets. For the 21 countries with positive adjustment requirements, the favourable combination lowers the DSA target by 0.36 percentage points of GDP on average and the adverse combination raises it by 0.65 points. The average required baseline adjustment is 2.6 points over four years. Per year, the two changes amount to about 0.09 and 0.16 points. The adverse change over four years is thus about as large as the cumulative threshold for deviations in the control account (Council of the European Union, 2024). Because I remove the average error from each series, the draws measure the spread of past forecast errors but not a persistent bias in the forecasts.

The stochastic criterion, which featured prominently in previous reform proposals (Blanchard et al., 2021), only sets the target for Finland. As a result, planned revisions of the stochastic DSA by the Commission, which would allow for persistent and non-normal shocks (Bec et al., 2025), affect average targets little. A VAR bootstrap that addresses both points raises the DSA target of the countries with adjustment requirements by 0.02 points on average. It changes the target substantially only for Finland.

The calibration of assumptions that determine how much risk policymakers are willing to tolerate has considerable effects on targets. These assumptions are the probability threshold of the stochastic criterion, the size of the stress tests and the period for covering rising ageing costs. I refer to these settings as the risk tolerance. Within the alternatives I consider, these settings can move the targets by amounts comparable to the uncertainty about forecasts and parameters. A strict combination of assumption increases target sby 0.18, while a lenient lowers it by 0.4. For the 21 countries with adjustment requirements, adverse interest-growth shocks of 0.75 instead of 0.5 percentage points raise the DSA target by 0.16 points. Comparing the shock size in the adverse $r - g$ test with past forecast errors shows that the baseline shock size of 0.5 percentage point is exceeded in 30 percent of cases. The test thus roughly corresponds to insuring against the 70 percent of cases in which forecasts overestimate $r - g$. Although this is the same level as the stochastic probability threshold, the $r - g$ stress test target exceeds the stochastic target in most cases.

Assumptions about the fiscal multiplier matter mostly through their interaction with the debt sustainability safeguard. The safeguard requires the debt ratio to decline by a minimum amount during the adjustment. A larger multiplier slows this decline, so that the binding targets of Finland, Austria and Germany rise with the multiplier. With an exceptionally high multiplier of 1.5, typically estimated for crisis conditions, Finland and Spain cannot meet the safeguard at all. The multiplier mainly delays the decline in debt, however. After the adjustment, debt falls at a similar pace whatever the multiplier.

A repayment profile based on Eurostat's residual debt maturity data primarily affects the

targets of Greece and Finland. Repeating the analysis with input data from September 2026 barely changes the average effect of each input, although the binding criterion changes for several countries.

The results have three implications for the review of the DSA. First, the size of the stress tests and the ageing cost period deserve as much attention as the stochastic simulations. Second, the debt sustainability safeguard's test of debt decline during the adjustment can introduce adverse feedback via the fiscal multiplier. Third, the level of risk that the rules implicitly accept via stress tests and probability thresholds should be justified explicitly.

The paper relates to a growing literature that evaluates the reformed EU rules and the role of the DSA in them (Heimberger, 2023; Pench, 2024, 2025; Cottarelli, 2024; Wieland, 2024; Jousten, 2025; Erce, 2025; Gros and Hofer, 2026; Boivin and Darvas, 2025; European Fiscal Board, 2025). In Darvas et al. (2023, 2024), we use a replication of the Commission's DSA to evaluate the Commission's proposal and the adopted rules. Several of these studies identify the inputs of the DSA as a source of uncertainty or discretion, but most do not quantify how much the targets depend on them. Other work, including our own, examines single inputs: the size and persistence of the fiscal multiplier (Heimberger et al., 2024), demographic risks (Darvas et al., 2025b), the growth assumptions in Germany's medium-term plan (Zettelmeyer et al., 2025), the specification of the stochastic DSA (Darvas et al., 2025a), and the effect of public investment and structural reforms on potential growth (Gottschalk et al., 2025). This paper complements these studies by quantifying the effects of a wider set of inputs in one framework, with ranges based on past forecast errors and published estimates.

The rest of the paper is organised as follows. Section 2 describes how the DSA sets the fiscal adjustment targets and classifies the inputs. Section 3 reports the effects of the forecasts and the model parameters, including the joint draws. Section 4 turns to the stochastic analysis, the stress tests and the calibration of risk tolerance, and compares their effects with those of the forecasts and parameters. Section 5 covers the repayment profile of debt and repeats the checks with newer data. Section 6 concludes and discusses channels that the DSA leaves out.

2 Fiscal adjustment targets

2.1 Debt sustainability analysis in the EU fiscal rules

The baseline of this sensitivity analysis is the replication of the Commission's DSA that we present in Darvas et al. (2024). The DSA model projects the public debt-to-GDP ratio using the standard debt-dynamics identity (European Commission, 2024b), in which nominal growth, the primary balance, and the average effective interest rate on debt depend on the SPB path through a fiscal multiplier and the budget semi-elasticity (Carnot and de Castro, 2015; Mourre

et al., 2019).[1] Annex 2 of Darvas et al. (2024) and the accompanying repository describe the methodology in detail. Because the input data for the baseline projection are taken from the Commission's 2024 prior guidance calculation sheets, the reference year is 2024 and the four-year adjustment spans the period from 2025 to 2028.

The DSA determines the target level for the SPB, defined as the lowest balance at the end of the adjustment period that satisfies a given set of criteria. For the purpose of the sensitivity analysis, I differentiate between the *DSA target*, which meets the criteria of the DSA alone, and the *binding target*, which is additionally compliant with the numerical safeguards and the EDP. The DSA itself comprises three types of criteria. First, under the deterministic debt criteria, the debt ratio must continuously decline, or stay below 60 percent of GDP, over the ten years following the adjustment period. This requirement must hold both in a baseline scenario and across three deterministic stress tests assuming a lower SPB, a higher interest-growth differential ($r - g$), and a temporary rise in market interest rates. Second, under the deficit criterion, the headline deficit must remain below 3 percent of GDP over the same ten-year post-adjustment window. Third, under the stochastic criterion, the debt ratio must decline over the five years following the adjustment period with a probability of at least 70 percent, which the model computes by simulating shocks to growth, interest rates, exchange rates, and the primary balance.

To obtain the binding target, the SPB path must satisfy four additional constraints beyond the DSA criteria: an adjustment floor described below, the minimum annual adjustment under the EDP, the debt sustainability safeguard, and the deficit resilience safeguard. The debt sustainability safeguard requires a minimum average decline in the debt ratio over the adjustment years outside the EDP, amounting to at least 1 percentage point of GDP per year when debt exceeds 90 percent of GDP, and at least 0.5 points per year when debt lies between 60 and 90 percent. The deficit resilience safeguard requires a minimum annual improvement in the structural balance until the structural deficit falls below 1.5 percent of GDP.

I implement these rules following the Commission's prior guidance calculation sheets, which include four noteworthy institutional features. First, the deterministic debt criteria apply in one of two variants across all scenarios simultaneously: either debt declines in the baseline and in all three stress tests (while also declining with at least 70 percent probability in the stochastic analysis), or debt remains below 60 percent of GDP in the baseline and in the three stress tests, with the less demanding variant applying. Second, the annual adjustment is held constant across the adjustment period, even though the regulation permits front-loaded adjustments that could, in interaction with the safeguards, reduce the total required adjustment in some cases.[2] Third, the annual adjustment is subject to an adjustment floor, meaning that required adjustment cannot be negative for countries receiving a reference trajectory. Fourth, countries with debt below 60 percent of GDP and a deficit below 3 percent receive only technical information; for these

[1] The budget semi-elasticity measures the change in the budget balance, in percent of GDP, when output changes by 1 percent.

[2] See the accompanying GitHub repository for a comparison of implementations with and without front-loading.

Table 1: Baseline SPB targets and Commission prior guidance, four-year adjustment (percent of GDP)

	2024		DSA target		Binding target			
	Debt	SPB	Comm.	Model	Comm.	Model	DSA criterion	Binding constraint
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Greece	153.9	1.73	2.37	2.33	2.61	2.58	Adverse $r-g$ test: debt declines	Deficit resilience
Italy	138.6	−1.05	3.35	3.31	3.35	3.31	Adverse $r-g$ test: debt declines	DSA criteria
France	112.4	−2.97	0.79	0.75	0.79	0.75	Adverse $r-g$ test: debt declines	DSA criteria
Spain	105.5	−0.85	2.59	2.59	2.59	2.59	Adverse $r-g$ test: debt declines	DSA criteria
Belgium	103.4	−2.03	1.25	1.21	1.25	1.21	Adverse $r-g$ test: debt declines	DSA criteria
Portugal	95.6	2.20	2.52	2.52	2.52	2.52	Adverse $r-g$ test: debt declines	DSA criteria
Finland	80.5	−0.53	2.11	1.83	5.87	4.83	Stochastic	Debt safeguard
Austria	79.5	−1.34	0.90	0.82	2.58	2.22	Adverse $r-g$ test: debt declines	Debt safeguard
Hungary	74.3	0.03	2.67	2.67	2.67	2.67	Adverse $r-g$ test: debt declines	DSA criteria
Cyprus	70.6	3.48	3.48	3.48	3.48	3.48	Adjustment floor	Adjustment floor
Slovenia	68.1	−1.24	0.52	0.52	0.52	0.52	Lower-SPB test: debt below 60% at end	DSA criteria
Germany	62.5	−1.02	1.14	1.02	2.74	2.46	Adverse $r-g$ test: debt below 60% at end	Debt safeguard
Croatia	59.5	−1.96	–	−0.44	−0.36	−0.36	Baseline: debt below 60% at end	Deficit resilience
Slovakia	58.5	−4.26	0.94	0.82	0.94	0.82	Lower-SPB test: debt below 60% at end	DSA criteria
Poland	53.7	−2.61	0.67	0.59	0.67	0.59	Deficit below 3%	DSA criteria
Malta	52.0	−2.90	−1.14	−1.22	−0.87	−0.87	Deficit below 3%	EDP minimum adjustment
Romania	50.9	−4.41	0.47	0.27	0.47	0.27	Deficit below 3%	DSA criteria
Netherlands	47.1	−0.64	–	0.08	0.08	0.08	Deficit below 3%	DSA criteria
Czechia	45.2	−0.09	–	0.39	0.39	0.39	Deficit below 3%	DSA criteria
Latvia	44.5	−1.36	–	−1.24	−0.14	−0.13	Deficit below 3%	Deficit resilience
Ireland	42.5	2.53	–	−0.63	–	−0.63	Deficit below 3%	DSA criteria
Lithuania	38.3	−0.37	–	0.27	0.27	0.27	Deficit below 3%	DSA criteria
Sweden	32.0	0.69	–	−1.75	−0.83	−0.74	Deficit below 3%	Deficit resilience
Luxembourg	27.1	0.10	–	−0.22	–	−0.22	Deficit below 3%	DSA criteria
Denmark	26.5	2.91	–	−1.09	−1.09	−1.09	Adjustment floor	Adjustment floor
Bulgaria	24.5	−2.09	–	−2.05	−0.87	−0.88	Deficit below 3%	Deficit resilience
Estonia	21.4	−0.29	−0.29	−0.29	−0.29	−0.29	Adjustment floor	Adjustment floor

Source: Author's calculations based on the European Commission's prior guidance calculation sheets of 2024.
Note: SPB at the end of the adjustment period (2028). Columns (3) and (5) are the Commission's targets. Column (3) is the SPB in 2024 plus four times the DSA-based annual adjustment. Countries receiving technical information report no DSA-based adjustment. Columns (4) and (6) are the model baseline. Columns (3), (4) and (7) follow the Commission's definition of the DSA target, which includes the adjustment floor and the rounding of the annual adjustment; in the rest of the paper, the DSA target excludes both. Column (7) is the DSA criterion that sets the DSA target. Column (8) is the constraint that sets the binding target: the DSA criteria, the adjustment floor, the debt sustainability safeguard, the deficit resilience safeguard or the minimum adjustment of the EDP. Countries are ordered by debt in 2024.

member states, only the deficit criterion and the requirement that debt stays below 60 percent in the baseline scenario apply, and their adjustment floor is set to -1 percentage point per year.

A final implementation detail concerns the fiscal multiplier and the closure of the output gap. During the adjustment period, the Commission's calculation sheets specify the output gap following a fiscal consolidation as the sum of the non-adjustment baseline output gap and two-thirds of the difference between the previous year's output gap and its baseline, after which the gap closes linearly over three years. I instead use a more intuitive formulation of the output gap closure, in which the deviation of the output gap from its baseline in any given year equals the cumulative sum of the contemporaneous and lagged fiscal multiplier effects (that is, the change in the SPB multiplied by the fiscal multiplier). For the baseline persistence of three years, the

change in the output gap in year t is the sum of the current year's multiplier effect, two-thirds of the effect in $t-1$, and one-third of the effect in $t-2$. This formulation makes varying the persistence parameter straightforward: for a four-year persistence, the output gap deviation sums the current effect down to one-fourth of the effect in $t-3$, and for a five-year persistence down to one-fifth in $t-4$. This choice does not qualitatively affect the results of the sensitivity analysis, and Table A6 in the Annex repeats the multiplier checks using the Commission's exact rule.

2.2 Replication of the Commission targets

Table 1 compares the baseline model output with the Commission's published targets. Across member states, the median absolute difference from the Commission's DSA targets is 0.04 percentage points, with the largest discrepancy of 0.28 points occurring for Finland. For the binding target, the median difference is 0.01 points, while the largest difference is 1.04 points, again for Finland, where the debt sustainability safeguard binds.[3] The alternative specification of the fiscal multiplier and output gap closure explained above accounts for the differences in the three countries where the debt sustainability safeguard binds. For Finland, Austria, and Germany, the baseline model yields binding targets of 4.83, 2.22, and 2.46 percent of GDP, compared with 5.87, 2.58, and 2.74 percent in the Commission's sheets. When implementing the Commission's output gap rule instead, the model produces binding targets of 5.99, 2.58, and 2.78 percent (Table A6), bringing all three values within 0.12 points of the Commission's figures. Section 3.3 confirms that the substantive conclusions regarding the fiscal multiplier do not depend on this specification.

2.3 Determining the binding criterion

Because each criterion implies its own minimum SPB, the overall DSA target is determined by the most demanding requirement. Figure 1 displays the SPB required by each DSA criterion and each safeguard or EDP constraint in the baseline. Excluding the adjustment floor, the deficit criterion sets the DSA target for 14 countries, the adverse $r-g$ stress test for nine, the lower-SPB stress test for two, the baseline debt criterion for one, and the stochastic criterion for Finland. For Cyprus, Denmark, and Estonia, all DSA criteria permit a lower SPB than in 2024, meaning that the adjustment floor sets their binding target. Under the Commission's definition used in Table 1, which incorporates the floor directly into the DSA target for these three countries, the deficit criterion sets the DSA target for 11 countries.

Figure 1 plots the deterministic debt criteria in the variant relevant to each member state. For Cyprus, Slovenia, Germany, Slovakia, Poland, Malta, and Romania, the second variant (keeping debt below 60 percent of GDP over the ten post-adjustment years) is less demanding and therefore applies, whereas countries receiving only technical information face solely the

[3] For this comparison, the minimum adjustment floor is included in both the DSA target and the binding target, whereas all subsequent analyses include this floor only in the binding target.

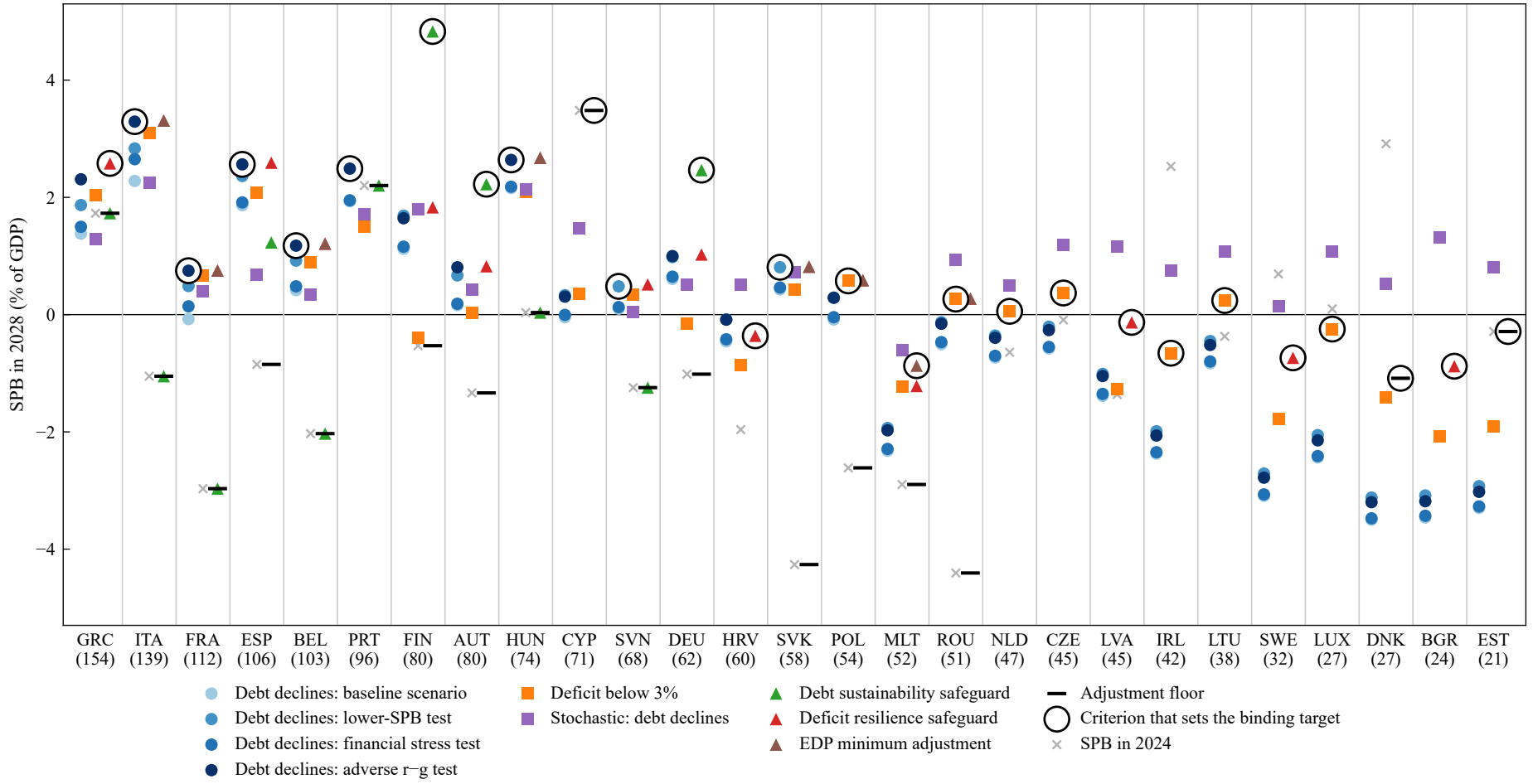


Figure 1: SPB required by each criterion in the baseline, four-year adjustment (percent of GDP)

Source: Author's calculations. Note: SPB at the end of the adjustment period (2028). Each column shows one country, ordered by debt in 2024 (in brackets), with the debt criteria (circles), the deficit and stochastic criteria (squares) and the constraints of the binding target (triangles) side by side. The open circle marks the criterion that sets the binding target. The bar shows the adjustment floor, for countries with a reference trajectory or where it binds. Grey crosses show the SPB in 2024. The debt criteria are shown in the variant that applies to the country. Section 2 defines the constraints.

deficit criterion and the baseline 60-percent debt ceiling. For the additional constraints of the binding target, the figure plots the standalone SPB required by the debt sustainability safeguard for countries with a reference trajectory and debt above 60 percent of GDP, while adding the minimum steps of the deficit resilience safeguard and the EDP to the DSA path for the countries where those rules apply. The adjustment floor equals the 2024 SPB for reference-trajectory countries and lies 4 percentage points below the 2024 SPB for technical-information countries. In the baseline, constraints outside the core DSA determine the binding target in 12 member states: the debt sustainability safeguard in Finland, Austria, and Germany; the deficit resilience safeguard in Greece, Croatia, Latvia, Sweden, and Bulgaria; the EDP minimum adjustment in Malta; and the adjustment floor in Cyprus, Denmark, and Estonia.

This maximum-operator structure implies that an input alters the DSA target only if it shifts the currently binding criterion or pushes a previously non-binding criterion above it, leaving changes to lower criteria without effect. As a consequence, favourable assumptions reduce the target only until the next criterion becomes binding, whereas adverse assumptions raise the required SPB without an upper bound. To evaluate the role of each assumption both in isolation and within the full regulatory framework, I report the sensitivity results at three levels: first, the effect on the input's *own criterion* (the specific stress test for stress-test shocks, the stochastic criterion for stochastic settings, and the baseline debt criterion for all macroeconomic forecasts and parameters); second, the effect on the overall DSA target; and third, the effect on the binding target.

2.4 Classification of the inputs

Table 2 organises the DSA inputs into four groups, detailing where each input enters the model, its baseline calibration, and the alternative values tested. The classification distinguishes between uncertainty surrounding the central debt projection itself, uncertainty regarding temporary shocks around that central path, and policy choices governing how much of both risks the fiscal rules tolerate.

The central projection corresponds to the baseline deterministic scenario and relies on medium-term forecasts for market interest rates, inflation, potential growth, and ageing costs, alongside structural parameters such as the fiscal multiplier and budget semi-elasticity. If these forecasts or parameter estimates are inaccurate, the entire baseline debt trajectory shifts, affecting every deterministic, stochastic, and safeguard criterion simultaneously. The first two input groups capture this baseline uncertainty by shifting the macroeconomic forecasts in line with historical forecast error distributions and varying the model parameters across the range of published empirical estimates.

The third group captures the estimation of temporary risk around the central projection through the stochastic analysis. In the Commission's framework, shock series are drawn from a joint normal distribution fitted to historical changes in growth, interest rates, exchange rates,

and the primary balance, and are added to the baseline paths to construct a fan chart of debt trajectories. Because these shocks have a zero mean and fade over time, the fan chart remains centred on the baseline projection and reflects year-to-year volatility rather than persistent forecast errors, such as overestimating ten-year potential growth. Methodological choices regarding the construction of this fan chart alter its width, skewness, and horizon, but leave the central projection and all deterministic criteria unchanged; consequently, they affect the fiscal targets solely through the stochastic criterion.

The fourth group encompasses the normative parameters that define the framework's risk tolerance: the probability threshold of the stochastic criterion, the shock sizes in the deterministic stress tests, and the horizon over which projected changes in ageing costs are incorporated into the SPB. The probability threshold dictates how much of the stochastic fan chart the debt reduction requirement must cover, the stress tests shift the central projection by fixed margins, and the ageing cost window determines the extent to which governments must pre-fund future demographic pressures rather than offsetting them later in the budget. Finally, the table includes a data check replacing the Commission's historical refinancing assumptions with Eurostat residual maturity data, which is examined in Section 5.

3 Forecasts and model parameters

3.1 Ranges of forecasts and model parameters

In the individual sensitivity checks, I shift each macroeconomic forecast by roughly half the interquartile range of its historical forecast error distribution: 1 percentage point for real interest rates and potential growth, 0.75 points for inflation, and 0.2 points of GDP for ageing costs. Applying symmetric shifts makes it easy to see whether fiscal targets respond more strongly to adverse errors than to favourable ones, whereas the joint simulations in Section 3.4 draw directly from the full empirical distributions. Throughout the analysis, a forecast error is defined as the realised outcome minus the forecast, pooled across EU countries. Table A1 in the Annex reports the underlying data sources and percentiles.

For real growth and inflation, I compute forecast errors from the autumn vintages of the IMF World Economic Outlook (WEO) between 2000 and 2018 (Ech-charfi, 2024), averaging the errors across the five forecast years for each country and vintage before demeaning by country. Prior to demeaning, growth was over-forecast by an average of 0.7 percentage points per year, reflecting both the secular decline in trend growth over the sample and the documented optimism bias of official euro-area projections (Frankel and Schreger, 2013). Because demeaning removes this average bias, the resulting error distributions capture the dispersion of forecast errors rather than systematic optimism in the baseline. The demeaned interquartile range spans from -0.84 to $+1.08$ percentage points for growth and from -0.74 to $+0.64$ points for inflation; because the WEO series measure actual rather than potential output growth and include recession episodes,

Table 2: Inputs of the DSA by group

Input	Where it enters	Baseline	Values checked
1. Forecasts: distribution of past forecast errors			
Real short-term rate, $T + 10$	Interest on new and refinanced debt	Commission	±1 pp.
Real long-term rate, $T + 10$	Interest on new and refinanced debt	Commission	±1 pp.
Inflation, $T + 10$	Nominal growth and rates	Commission	±0.75 pp.
Potential growth, $T + 10$	Real growth	Commission	±1 pp.
Ageing costs, change 2024–2038	SPB after the adjustment	Ageing Report	±0.2 pp. of GDP
Rates and inflation, $T + 30$	Long-run rates and growth	Commission	±0.5 and ±0.25 pp.[a]
2. Model parameters: range of published estimates			
Fiscal multiplier	Output effect of the adjustment	0.75	0.5 to 1.0; 1.5[b]
Persistence of the multiplier	Output effect of the adjustment	3 years	2 to 5 years
Budget semi-elasticity	Cyclical balance	Commission	×0.9 and ×1.1
3. Estimation of risk: alternative choices			
Start of stochastic projection	Stochastic criterion	After adjustment	First adjustment year
Horizon of stochastic projection	Stochastic criterion	5 years	10 years
Start of shock sample	Stochastic criterion	2000	1990, 2010
Shock frequency	Stochastic criterion	Quarterly	Annual
Shock distribution	Stochastic criterion	Normal	VAR bootstrap
Winsorisation of shocks	Stochastic criterion	5th/95th pct.	1st/99th pct., none
4. Risk tolerance: alternative settings			
Probability threshold	Stochastic criterion	70%	60%, 80%
Adverse $r - g$ shock	Stress test	0.5 pp.	0.25, 0.75 pp.
Financial stress shock	Stress test	1 pp.	0.5, 1.5 pp.
Lower-SPB shock	Stress test	0.5 pp.	0.25, 0.75 pp.
Ageing cost period	SPB after the adjustment	10 years	5 years
Data			
Repayment profile of debt	Refinancing	Historical average	Eurostat residual maturity

Note: Values of the forecasts are deviations from the baseline, in percentage points (pp.), of about the size of typical past errors; the joint draws use the full distribution of past errors. Forecasts and model parameters are varied one at a time and drawn jointly. All other inputs are varied one at a time. Table A1 in the Annex reports the sources of the forecast errors. [a] Interest rates ±0.5 pp., inflation ±0.25 pp.; individual checks only, since there is no evidence on errors at this horizon. [b] Individual check only, for crisis conditions.

this range represents a conservative upper bound on uncertainty around medium-term potential growth.

For interest rates, the DSA anchors market expectations ten years ahead using forward rates. Because Germany is the only member state with a public zero-coupon yield curve extending over a sufficiently long historical sample, I evaluate forward-rate errors using year-end data from the Bundesbank between 1986 and 2015, comparing ten-year-ahead forward rates with the spot rates realised a decade later. Raw forward rates exceeded realised ten-year yields by an average of 3.0 percentage points due to positive term premia and the secular decline in nominal

rates since the 1990s. After demeaning to remove this trend, the interquartile range runs from −1.24 to +0.89 percentage points for the long-term rate and from −1.22 to +0.40 points for the short-term rate. Relying on the German curve omits forecast errors in sovereign spreads over Bunds, which are largest for high-debt countries such as Italy and Greece and thus make these ranges a lower bound on interest-rate uncertainty for high-spread issuers. Conversely, applying nominal forward-rate errors directly to real rates while shocking inflation independently makes the implied dispersion of nominal rates slightly wider than in the historical data.

To calibrate uncertainty around ageing costs, I use the eight sensitivity scenarios of the 2024 Ageing Report, which vary demographic, labour-market, and healthcare cost assumptions (European Commission, 2024a). Taking the deviation of each scenario from the baseline change in ageing-related expenditure between 2024 and 2038 (the end of the DSA horizon for a four-year adjustment) yields an interquartile range of −0.24 to +0.14 percentage points of GDP.

Letting T denote the base year of the projection (2024 in the baseline), the shifts to real interest rates, inflation, and potential growth are applied at the $T + 10$ anchor. Each shift phases in linearly from zero in the final forecast year to its full value at $T + 10$, and then fades linearly back to zero by $T + 30$ in accordance with the Commission's interpolation scheme between medium- and long-run anchors. Real GDP growth moves one-for-one with potential growth so that the output gap path is unaffected, while nominal interest rates move one-for-one with inflation shocks because the rate anchors are specified in real terms. The ageing-cost shock accumulates linearly from 2024 until reaching its full magnitude in 2038.

The parameter ranges are drawn from the empirical literature. The baseline fiscal multiplier is 0.75, following Carnot and de Castro (2015). Because standard estimates of government spending multipliers in normal times lie between 0.6 and 1.0 (Ramey and Zubairy, 2018; Ramey, 2019), with Barro and Redlick (2011) finding impact multipliers of 0.4 to 0.5, I vary the multiplier between 0.5 and 1.0 in both the individual checks and the joint draws. Substantially larger multipliers of around 1.5 have been documented during the post-2010 European consolidations, when monetary policy was constrained by the effective lower bound and fiscal retrenchment occurred simultaneously across countries (Blanchard and Leigh, 2013), as well as in cross-regional studies of currency unions (Nakamura and Steinsson, 2014; Chodorow-Reich, 2019), although regional estimates reflect externally financed spending rather than budget-neutral national consolidations. To capture severe downturn or zero-lower-bound episodes without distorting the normal-times distribution, I evaluate a crisis multiplier of 1.5 as a separate individual check rather than including it in the joint draws. Furthermore, because cumulative multipliers over two years are at least as large as impact multipliers (Barro and Redlick, 2011; Ramey and Zubairy, 2018) and fiscal consolidations can exert persistent hysteresis effects on output (Fatás and Summers, 2018; Gechert et al., 2019), I vary the persistence of the multiplier effect from two to five years around the three-year baseline. Finally, I scale the budget semi-elasticity by factors of 0.9 and 1.1, matching the largest country revisions of roughly 10 percent observed between the Commission's 2014 and 2018 parameter updates (Mourre et al., 2019).

Table 3a: Change in the SPB relative to the baseline: forecasts and model parameters, four-year adjustment (percentage points of GDP)

		Own criterion		DSA target		Binding target	
Input	Value	Mean	Largest	Mean	Largest	Mean	Largest
1. Forecasts							
Real short-term rate, $T + 10$	−1 pp.	−0.04	−0.15 ITA	−0.04	−0.14 ITA	−0.03	−0.12 PRT
	+1 pp.	+0.04	+0.15 ITA	+0.04	+0.14 ITA	+0.03	+0.16 ITA
Real long-term rate, $T + 10$	−1 pp.	−0.26	−0.52 GRC	−0.23	−0.52 GRC	−0.17	−0.44 ESP
	+1 pp.	+0.25	+0.53 GRC	+0.23	+0.47 GRC	+0.17	+0.40 ITA
Inflation, $T + 10$	−0.75 pp.	+0.18	+0.46 ITA	+0.11	+0.45 ITA	+0.10	+0.44 ITA
	+0.75 pp.	−0.16	−0.45 ITA	−0.04	−0.26 FIN	−0.03	−0.20 HUN
Potential growth, $T + 10$	−1 pp.	+0.54	+1.23 ITA	+0.42	+1.21 ITA	+0.35	+1.20 ITA
	+1 pp.	−0.51	−1.18 ITA	−0.29	−0.65 ESP	−0.20	−0.64 ESP
Ageing costs	−0.2 pp.	−0.09	−0.15 AUT	−0.08	−0.14 AUT	−0.06	−0.16 BEL
	+0.2 pp.	+0.09	+0.15 AUT	+0.08	+0.14 AUT	+0.06	+0.12 ESP
Real short-term rate, $T + 30$	−0.5 pp.	0.00	−0.01 PRT	0.00	−0.01 PRT	0.00	−0.04 PRT
	+0.5 pp.	0.00	+0.01 PRT	0.00	+0.01 PRT	0.00	0.00
Real long-term rate, $T + 30$	−0.5 pp.	−0.01	−0.03 GRC	−0.01	−0.02 GRC	−0.01	−0.04 PRT
	+0.5 pp.	+0.01	+0.03 GRC	+0.01	+0.02 GRC	0.00	+0.03 GRC
Inflation, $T + 30$	−0.25 pp.	+0.01	+0.04 GRC	+0.01	+0.04 GRC	0.00	+0.03 GRC
	+0.25 pp.	−0.01	−0.04 GRC	−0.01	−0.04 GRC	−0.01	−0.04 PRT
2. Model parameters							
Fiscal multiplier	0.5	−0.01	−0.07 SVK	−0.04	−0.25 POL	−0.13	−1.00 FIN
	1	+0.01	+0.07 SVK	+0.06	+0.38 FRA	+0.22	+1.64 FIN
	1.5	+0.04	+0.22 SVK	+0.28	+1.77 FRA	+0.95	+8.56 AUT
Multiplier persistence	2 years	−0.01	−0.05 SVK	−0.03	−0.18 POL	−0.08	−0.64 FIN
	4 years	+0.01	+0.06 SVK	+0.04	+0.24 FRA	+0.10	+0.68 FIN
	5 years	+0.03	+0.13 FIN	+0.08	+0.45 FRA	+0.18	+1.20 FIN
Budget semi-elasticity	×0.9	−0.01	−0.03 DEU	−0.01	−0.08 POL	−0.05	−0.40 FIN
	×1.1	+0.01	+0.03 DEU	+0.01	+0.08 POL	+0.06	+0.40 FIN

Source: Author's calculations. Note: Mean of the change across the 21 countries with adjustment requirements, and the largest change in absolute terms among them with its country. Own criterion: SPB required by the criterion the check acts on (Section 2). Runs in which the debt sustainability safeguard cannot be met are left out of the binding target. These are the runs with a multiplier of 1.5 for Finland and Spain.

3.2 Individual variation of inputs

In the 2024 baseline, 21 member states face a positive adjustment requirement, meaning that their binding SPB target exceeds their initial 2024 SPB; all cross-country summary statistics in the remainder of the paper refer to this group of 21 countries. Tables 3a and 3b report the mean and maximum changes in the required SPB across these countries at all three target levels for a four-year adjustment period, while Tables A2a and A2b in the Annex present the corresponding results for a seven-year adjustment.

Among the macroeconomic forecasts, medium-term potential growth and long-term interest rates exert the strongest influence on the fiscal targets. Lowering potential growth at $T + 10$ by 1 percentage point raises the DSA target by an average of 0.42 percentage points of GDP, with the largest increases occurring in Italy (1.21 points), France (1.05 points), and Greece (0.90 points). By contrast, raising potential growth by 1 percentage point reduces the average DSA target by only 0.29 points, and by at most 0.65 points (in Spain). This asymmetry arises

Table 3b: Change in the SPB relative to the baseline: estimation of risk, risk tolerance and data, four-year adjustment (percentage points of GDP)

		Own criterion		DSA target		Binding target	
Input	Value	Mean	Largest	Mean	Largest	Mean	Largest
3. Estimation of risk							
Start of stochastic projection	first adjustment year	+0.20	+0.87 GRC	0.00	+0.09 FIN	0.00	0.00
Horizon of stochastic projection	10 years	+0.14	+0.69 GRC	−0.01	−0.11 FIN	0.00	0.00
Start of shock sample	1990	−0.02	−0.08 GRC	0.00	0.00	0.00	0.00
	2010	+0.01	+0.35 SVN	0.00	+0.09 FIN	0.00	0.00
Shock frequency	annual	−0.14	−0.55 MLT	0.00	+0.07 FIN	0.00	0.00
Shock distribution	VAR bootstrap	+0.06	+0.70 GRC	+0.02	+0.46 FIN	0.00	0.00
Winsorised shocks	P1 and P99	+0.52	+1.11 ESP	+0.04	+0.36 FRA	+0.02	+0.36 FRA
	off	+0.83	+1.70 ESP	+0.09	+0.63 AUT	+0.04	+0.64 FRA
4. Risk tolerance							
Threshold, start, horizon	60%, after adj., 5 years	−0.49	−0.77 GRC	−0.01	−0.11 FIN	0.00	0.00
	80%, after adj., 5 years	+0.58	+1.05 GRC	+0.03	+0.46 FIN	+0.01	+0.08 FRA
	60%, after adj., 10 years	−0.26	−0.93 FIN	−0.01	−0.11 FIN	0.00	0.00
	80%, after adj., 10 years	+0.59	+1.61 GRC	+0.03	+0.58 GRC	+0.02	+0.45 GRC
	60%, first adj. year, 5 years	−0.42	−0.71 MLT	−0.01	−0.11 FIN	0.00	0.00
	80%, first adj. year, 5 years	+0.93	+2.51 GRC	+0.15	+1.48 GRC	+0.09	+1.23 GRC
	60%, first adj. year, 10 years	−0.17	−0.91 FIN	−0.01	−0.11 FIN	0.00	0.00
	70%, first adj. year, 10 years	+0.32	+1.56 GRC	+0.02	+0.54 GRC	+0.02	+0.39 GRC
	80%, first adj. year, 10 years	+0.89	+3.03 GRC	+0.13	+2.00 GRC	+0.11	+1.75 GRC
Adverse $r - g$ shock	0.25 pp.	−0.28	−0.50 ITA	−0.06	−0.27 GRC	−0.05	−0.28 BEL
	0.75 pp.	+0.28	+0.49 ITA	+0.16	+0.49 ITA	+0.12	+0.48 ITA
Lower-SPB shock	0.25 pp.	−0.23	−0.28 FIN	0.00	−0.05 SVK	−0.01	−0.08 SVN
	0.75 pp.	+0.23	+0.28 FIN	+0.06	+0.19 SVN	+0.04	+0.20 SVK
Financial stress shock	0.5 pp.	−0.02	−0.06 FRA	0.00	0.00	0.00	0.00
	1.5 pp.	+0.02	+0.10 FRA	0.00	0.00	0.00	0.00
Ageing cost period	5 years	−0.22	−0.93 ESP	−0.24	−0.89 ESP	−0.20	−0.88 ESP
All settings	lenient	−0.22	−0.93 ESP	−0.40	−1.25 ESP	−0.27	−1.28 ESP
	strict	0.00	0.00	+0.18	+0.49 ITA	+0.13	+0.48 ITA
Data							
Repayment profile	Eurostat profile	−0.01	−0.46 GRC	−0.01	−0.57 GRC	0.00	−0.42 GRC

Source: Author's calculations. Note: Mean of the change across the 21 countries with adjustment requirements, and the largest change in absolute terms among them with its country. Own criterion: SPB required by the criterion the check acts on (Section 2). Runs in which the debt sustainability safeguard cannot be met are left out of the binding target. The own criterion of the settings of the stochastic analysis and of the stress tests applies only to countries with a reference trajectory, and its statistics cover the 15 of them with adjustment requirements. Threshold, start, horizon: probability threshold of the stochastic criterion, start of the stochastic projection (after the adjustment period or in its first year) and horizon after the adjustment period. The baseline has a threshold of 70 percent, a start after the adjustment period, a five-year horizon, shocks of 0.5, 1 and 0.5 percentage points in the adverse $r - g$, financial stress and lower-SPB tests and an ageing cost period of ten years. Lenient: threshold of 60 percent, shocks of 0.25, 0.5 and 0.25 points and ageing cost period of five years. Strict: threshold of 80 percent and shocks of 0.75, 1.5 and 0.75 points. Hungary, Luxembourg and the Netherlands are left out of the repayment profile check for lack of data.

from the interaction of multiple criteria rather than from the debt dynamics themselves: on the baseline debt criterion alone, the required SPB responds almost symmetrically, rising by 0.54 points under lower growth and falling by 0.51 points under higher growth, but favourable growth assumptions quickly cause the deficit criterion to become the binding constraint.

Shifting the real long-term interest rate at $T + 10$ by 1 percentage point moves the DSA target symmetrically by 0.23 percentage points on average, raising the target by more than 0.1 points in 18 of the 21 countries and by up to 0.47 points in Greece, whereas shocks to the short-term interest rate have virtually no effect. Holding real rates constant, inflation shocks matter for only a few high-debt countries with long average debt maturities: a 0.75 percentage point reduction

in inflation raises the DSA target by 0.45 points in Italy, 0.43 points in France, and 0.30 points in Greece by slowing the nominal erosion of existing fixed-rate debt. Finally, variations in ageing costs and in the $T + 30$ long-run anchors have minor effects: a 0.2 percentage point shift in ageing costs by 2038 alters the average DSA target by 0.08 points (and by no more than 0.16 points in any country), reflecting the narrow dispersion across Ageing Report scenarios over this horizon, while shifting the $T + 30$ anchors changes no target by more than 0.04 points.

Varying the structural model parameters within normal-times ranges also has a modest impact on the DSA target. Increasing the fiscal multiplier from 0.75 to 1.0 raises the DSA target by an average of 0.06 percentage points (and at most 0.38 points), while varying the multiplier persistence or the budget semi-elasticity changes the average target by less than 0.1 points. Under a crisis multiplier of 1.5, however, the average DSA target rises by 0.28 points, driven by substantial increases in France (1.77 points), Poland (1.11 points), Romania (0.86 points), and Italy (0.75 points). In these member states, the deficit criterion either binds in the baseline or lies close to the binding threshold; a higher multiplier deepens the negative output gap during the consolidation period, which widens the cyclical deficit and requires a higher structural balance to keep the headline deficit within 3 percent of GDP.

3.3 Fiscal multiplier and debt sustainability safeguard

While the fiscal multiplier has a limited effect on the core DSA criteria in most countries, it interacts strongly with the debt sustainability safeguard when determining the binding target. Raising the multiplier from 0.75 to 1.0 increases the binding target by 1.6 percentage points in Finland, 1.0 point in Austria, and 0.8 points in Germany, while extending the multiplier persistence to five years raises Finland's binding target by 1.2 points. Under the crisis multiplier of 1.5, the binding target jumps by 8.6 percentage points in Austria and 4.1 points in Germany, while Finland and Spain cannot satisfy the debt sustainability safeguard even with an annual SPB adjustment of 5 percentage points (the upper bound of the numerical grid in the main runs, which requires omitting both countries from the mean binding target change in Table 3a).

This feedback loop arises because the debt sustainability safeguard requires a minimum average reduction in the debt-to-GDP ratio during the adjustment period itself, a window over which a larger multiplier depresses nominal GDP and slows debt reduction (Heimberger et al., 2024). A stylised impact calculation ignoring interest rates, inflation, and lagged output effects illustrates the mechanism (see also Boussard et al., 2012; Eyraud and Weber, 2013). Letting m denote the fiscal multiplier, ε the budget semi-elasticity, and b the initial debt-to-GDP ratio, an SPB consolidation of x percentage points improves the headline balance by $x(1 - \varepsilon m)$ while raising the debt ratio by approximately bmx through the denominator effect on GDP. On impact, fiscal consolidation reduces the debt ratio only if $m(b + \varepsilon) < 1$; for a debt ratio of 80 percent of GDP and a semi-elasticity of 0.5, this condition requires a multiplier below 0.77. Over a multi-year adjustment period, the safeguard can still be met at somewhat higher multipliers

because the output contraction fades while the higher primary balance continues to pay down debt, but when the multiplier is both large and persistent, the denominator effect dominates for too long.

Panel (a) of Figure 2 traces the minimum constant annual adjustment needed to satisfy the debt sustainability safeguard as a function of the fiscal multiplier for the four countries where the safeguard binds or fails, expanding the grid search up to an annual adjustment of 20 percentage points in steps of 0.05 points. At the baseline multiplier of 0.75, the safeguard requires annual adjustments between 0.55 and 1.35 percentage points. The required effort increases gradually for multipliers up to about 1.2 but escalates nonlinearly thereafter: for Finland, the required annual adjustment reaches 4.2 percentage points at a multiplier of 1.44, and above 1.45 no adjustment up to 20 points per year can satisfy the safeguard; for Spain, the requirement reaches 4.0 points per year at a multiplier of 1.48 and becomes infeasible at 1.49. In Spain, the average annual debt reduction peaks below the required 1 percentage point threshold regardless of how aggressively the government tightens, whereas in Finland a larger consolidation depresses GDP enough in the early years to push the debt ratio above the 90 percent threshold, triggering the stricter 1-point annual debt reduction requirement. The infeasibility of the safeguard at high multipliers is thus an inherent property of the rule rather than an artefact of the 5-point search cap used in the main tables.

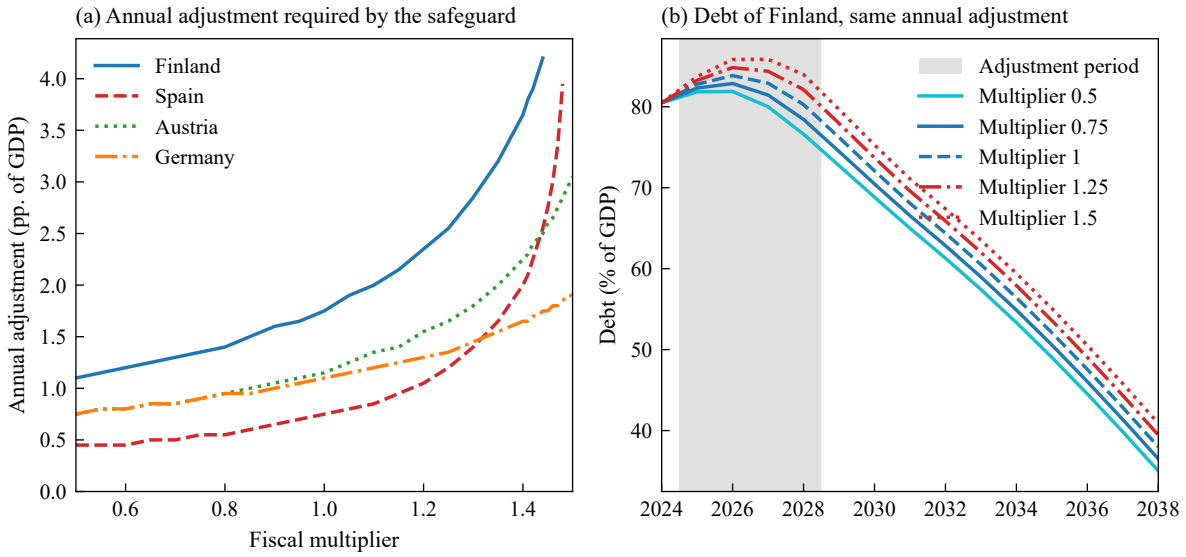


Figure 2: Debt sustainability safeguard and fiscal multiplier, four-year adjustment

Source: Author's calculations. Note: (a) Smallest constant annual adjustment, in steps of 0.05 percentage points, for which the debt sustainability safeguard is met, including the minimum adjustment of the EDP and the deficit resilience safeguard, with the persistent multiplier of the baseline, for multipliers from 0.5 to 1.5, in steps of 0.05 and of 0.01 above 1.4. Lines end where no annual adjustment of up to 20 percentage points meets the safeguard. The baseline multiplier is 0.75. (b) Debt of Finland for a constant annual adjustment of 1.34 percentage points, the adjustment of its baseline binding target. The safeguard is met with multipliers of 0.5 and 0.75 and not met with multipliers of 1.0 and more. The shaded area marks the adjustment period.

Crucially, the fiscal multiplier alters the timing of debt reduction much more than its long-run trajectory. Panel (b) of Figure 2 plots Finland's debt ratio under a fixed annual adjustment of

1.34 percentage points (its baseline binding requirement) for multipliers ranging from 0.5 to 1.5. Starting from 80.5 percent of GDP in 2024, the 2028 debt ratio at the end of the adjustment period reaches 76.6 percent under a multiplier of 0.5, compared with 84.0 percent under a multiplier of 1.5, violating the safeguard for all multipliers of 1.0 or above. Once the adjustment period ends and the output gap closes, however, the debt ratio declines at a nearly identical pace of around 4 percentage points per year across all multiplier values, reaching 35.0 percent of GDP in 2038 under a multiplier of 0.5 and 40.9 percent under a multiplier of 1.5 (with the level difference reflecting the higher deficits accumulated during the adjustment window). Because the debt sustainability safeguard evaluates debt reduction strictly during the adjustment years, it is highly sensitive to temporary output effects; evaluating debt reduction over a longer post-adjustment window would avoid requiring pro-cyclical tightening.

Although the long-run debt impact of a temporary multiplier is modest, short-run self-defeating dynamics remain a policy concern if large adjustments are required during a downturn (Boussard et al., 2012; Eyraud and Weber, 2013; Heimberger et al., 2024). Under a multiplier of 1.25 or higher, Finland's debt ratio rises to roughly 85 percent of GDP during the first two adjustment years and ends the four-year period above its 2024 starting level, meaning that aggressive consolidation initially worsens headline debt metrics. Empirical evidence from the post-2010 European consolidations associates multipliers of this magnitude with periods in which monetary policy is constrained at the effective lower bound and multiple trading partners consolidate simultaneously (Blanchard and Leigh, 2013; House et al., 2020), or when consolidations are undertaken during recessions (Jordà and Taylor, 2016), potentially inducing persistent output losses (Fatás and Summers, 2018; Gechert et al., 2019), even if expenditure-based consolidations in normal times generally entail smaller output costs (Guajardo et al., 2014; Alesina et al., 2019). Because the joint draws in Section 3.4 restrict the multiplier to the normal-times interval between 0.5 and 1.0, the safeguard remains feasible across all 500 draws, confirming that safeguard infeasibility is specifically a crisis-regime risk.

In Finland, the stringency of the debt sustainability safeguard is further amplified by the Commission's stock-flow adjustment projections, which reflect public pension fund surpluses that improve the general government balance without reducing gross public debt (Boivin and Darvas, 2025; Darvas et al., 2024). Following the baseline replication, I retain these country-specific projections across all simulations. Finally, Table A6 in the Annex shows that replacing the baseline output gap rule with the Commission's exact specification yields even larger multiplier effects: under a multiplier of 1.5, Austria also fails to meet the safeguard and Germany's binding target rises by 7.00 rather than 4.08 percentage points, confirming that the safeguard vulnerability is robust to the output gap formulation.

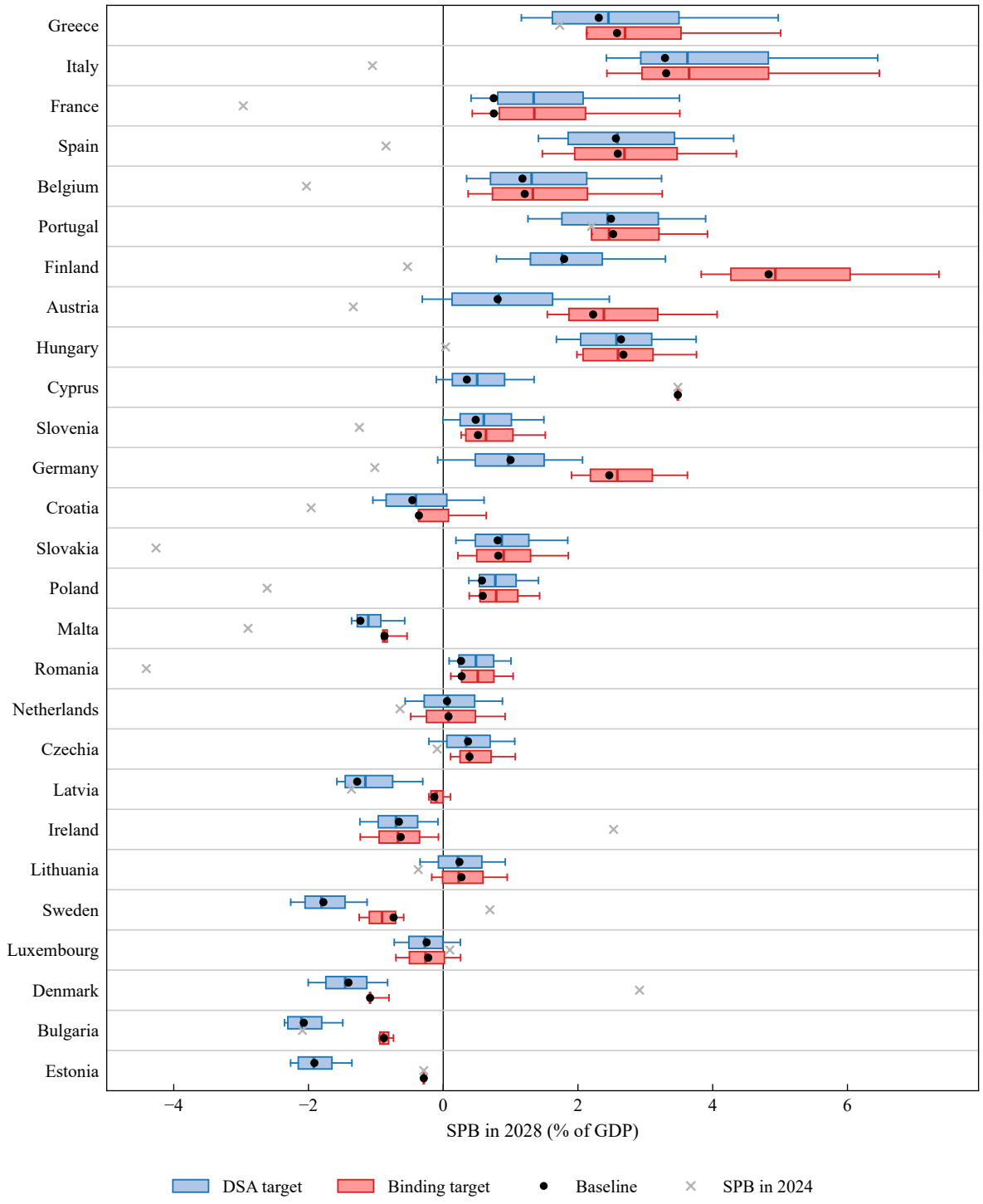


Figure 3: SPB targets by country: baseline and joint draws, four-year adjustment (percent of GDP)

Source: Author's calculations. Note: SPB at the end of the adjustment period. For each country, boxes show the 25th to 75th percentiles of the DSA target (blue) and of the binding target (red) across 500 joint draws of forecasts and model parameters, lines in the boxes the median and whiskers the 10th to 90th percentiles. Dots show the baseline targets and grey crosses the SPB in 2024. Countries are ordered by debt in 2024.

3.4 Joint variation of forecasts and model parameters

Because varying inputs one at a time misses compounding interactions (such as a high fiscal multiplier combined with longer persistence, or simultaneous adverse shocks that push multiple criteria above the baseline target), I next conduct 500 joint draws of all forecasts and model parameters using Latin hypercube sampling. Forecast errors are drawn from their empirical distributions, pairing variables that come from the same underlying dataset: short- and long-term interest rate errors from the same calendar year are drawn together, as are growth and inflation errors from the same country and WEO vintage. Errors from different datasets are drawn independently because their sample periods and forecast horizons differ; since growth and interest rate errors were positively correlated during the overlapping years from 2000 to 2015, independent sampling slightly overstates the variance of $r - g$. For the structural parameters, the fiscal multiplier and budget semi-elasticity are drawn from uniform distributions over $[0.5, 1.0]$ and $[0.9, 1.1]$ times the baseline, respectively, while multiplier persistence takes integer values from two to five years with equal probability. All stochastic methodology and risk-tolerance parameters remain at their baseline settings, and each draw executes a full run of the DSA model, including the stochastic simulations.

To summarise the resulting target distributions without giving undue weight to extreme historical outliers (such as the 2006–2008 boom-bust forecast errors in Ireland, Greece, and the Baltic states), I focus on the median and the interquartile range across the 500 draws, referring to the 25th percentile as the favourable combination of assumptions and the 75th percentile as the adverse combination. Figure 3 and Table 4 present the country-level distributions of the DSA and binding targets, while Figure A1 in the Annex illustrates the fan of debt trajectories when countries follow their baseline binding SPB path across the 500 parameter and forecast draws. Across the 21 member states with positive adjustment requirements (six of which receive only technical information), the baseline binding target requires an average SPB improvement of 2.6 percentage points of GDP over four years. Relative to the baseline, the favourable combination lowers the average DSA target by 0.36 percentage points, the median draw raises it by 0.09 points, and the adverse combination raises it by 0.65 points (corresponding to annual adjustment shifts of −0.09 and +0.16 points per year). For the binding target, the favourable combination reduces the required SPB by 0.26 points on average, the median raises it by 0.10 points, and the adverse combination raises it by 0.63 points.

To put these magnitudes in perspective, the reformed rules monitor compliance through a control account that records deviations of net expenditure from the agreed path, triggering an Article 126(3) report for countries with debt above 60 percent of GDP if deviations exceed 0.3 percentage points of GDP in a single year or 0.6 points cumulatively (Council of the European Union, 2024). Although net expenditure and the SPB are not identical, the 0.65 percentage point increase in the four-year SPB target under the adverse combination is roughly equivalent to the entire cumulative deviation permitted by the control account.

As in the individual checks, the target distributions are skewed toward higher requirements because adverse draws push the binding criterion upward without bound, whereas favourable draws are truncated whenever another DSA criterion or the adjustment floor becomes binding. Notably, this asymmetry does not stem from the underlying forecast error distributions, which between the 25th and 75th percentiles are slightly wider on the favourable side for both interest rates and growth, with large adverse interest-rate errors appearing only around the 90th percentile (Table A1).

Across individual member states, the interquartile band of the DSA target is widest for Italy, spanning from 0.36 percentage points below the baseline to 1.53 points above it, followed by Greece, Spain, Austria, Portugal, and Belgium with interquartile ranges between 1.4 and 1.9 points. The upward skewness is most pronounced in France, Poland, and Romania, where the deficit criterion binds or nearly binds in the baseline: in these countries, draws with a multiplier above 0.75 raise the SPB required by the deficit criterion while favourable macro draws quickly hit competing criteria, causing even the 25th percentile draw in France to exceed the baseline DSA target by 0.06 points (while the 75th percentile raises it by 1.33 points). In most other member states, the dispersion is driven primarily by the macroeconomic forecasts rather than the multiplier, as a multiplier of 1.0 shifts the standalone DSA target by more than 0.1 points in only four countries.

When moving from the DSA target to the binding target, the debt sustainability safeguard amplifies the upper tail in Finland, Austria, and Germany. At the 75th percentile, Finland's binding target rises by 1.21 percentage points (compared with 0.56 points for its DSA target), while the binding targets of Austria and Germany rise by 0.96 and 0.64 points, respectively. Across all 21 countries and 500 draws, 11 percent of country-draws require a cumulative four-year SPB adjustment exceeding 5 percentage points of GDP, a consolidation size that Darvas et al. (2025a) find occurred in fewer than 10 percent of all multi-year episodes across 36 advanced economies between 1970 and 2022. The share of draws requiring adjustments above 5 points of GDP reaches 66 percent in Finland and 59 percent in Slovakia (where the baseline target already exceeds 5 points), followed by 43 percent in Italy, 37 percent in Romania, and 26 percent in France.

4 Stochastic analysis, stress tests and risk tolerance

4.1 Specification of the stochastic analysis

The third group of inputs concerns the statistical specification used to estimate uncertainty in the stochastic criterion. I evaluate six methodological variations one at a time: first, starting the simulated shocks in the first adjustment year rather than after the adjustment period (while keeping primary balance shocks at zero during the adjustment years and still testing whether debt five years after the adjustment period lies below its end-of-adjustment level); second, extending

Table 4: SPB targets by country: baseline and joint draws of forecasts and model parameters, four-year adjustment (percent of GDP and percentage points)

			Own	DSA target					Binding target				
	Debt	Required	Band	Base	P25	Median	P75	Band	Base	P25	Median	P75	Band
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
Greece	153.9	0.85	2.48	2.31	−0.69	+0.14	+1.19	1.88	2.58	−0.45	+0.12	+0.95	1.40
Italy	138.6	4.36	2.79	3.29	−0.36	+0.33	+1.53	1.89	3.31	−0.36	+0.34	+1.52	1.88
France	112.4	3.72	1.89	0.75	+0.06	+0.59	+1.33	1.27	0.75	+0.08	+0.60	+1.36	1.28
Spain	105.5	3.44	1.93	2.56	−0.71	+0.02	+0.87	1.58	2.59	−0.64	+0.10	+0.88	1.52
Belgium	103.4	3.24	2.08	1.18	−0.47	+0.14	+0.95	1.43	1.21	−0.48	+0.12	+0.93	1.41
Portugal	95.6	0.32	1.54	2.49	−0.73	−0.05	+0.70	1.43	2.52	−0.32	−0.06	+0.68	1.00
Finland	80.5	5.36	1.05	1.79	−0.50	−0.03	+0.56	1.07	4.83	−0.56	+0.10	+1.21	1.77
Austria	79.5	3.56	1.67	0.81	−0.68	+0.02	+0.82	1.49	2.22	−0.36	+0.16	+0.96	1.32
Hungary	74.3	2.64	1.32	2.64	−0.60	−0.07	+0.46	1.06	2.67	−0.60	−0.08	+0.44	1.04
Cyprus	70.6	0.00	0.95	0.35	−0.22	+0.15	+0.56	0.78	3.48	0.00	0.00	0.00	0.00
Slovenia	68.1	1.76	0.88	0.48	−0.23	+0.12	+0.53	0.76	0.52	−0.18	+0.12	+0.52	0.70
Germany	62.5	3.48	1.02	1.00	−0.52	−0.03	+0.50	1.02	2.46	−0.28	+0.12	+0.64	0.92
Croatia	59.5	1.60	0.97	−0.46	−0.39	+0.05	+0.51	0.90	−0.36	0.00	0.00	+0.44	0.44
Slovakia	58.5	5.08	0.89	0.81	−0.33	+0.06	+0.46	0.79	0.82	−0.32	+0.08	+0.48	0.80
Poland	53.7	3.20	0.96	0.58	−0.04	+0.20	+0.51	0.54	0.59	−0.04	+0.20	+0.52	0.56
Malta	52.0	2.03	0.93	−1.23	−0.05	+0.12	+0.30	0.35	−0.87	−0.03	0.00	+0.04	0.07
Romania	50.9	4.68	0.91	0.26	−0.03	+0.22	+0.48	0.52	0.27	0.00	+0.24	+0.48	0.48
Netherlands	47.1	0.72	0.90	0.06	−0.34	+0.01	+0.41	0.75	0.08	−0.33	0.00	+0.40	0.73
Czechia	45.2	0.48	0.83	0.37	−0.32	−0.03	+0.33	0.64	0.39	−0.14	0.00	+0.32	0.46
Latvia	44.5	1.23	0.89	−1.28	−0.18	+0.12	+0.53	0.70	−0.13	−0.05	+0.03	+0.13	0.18
Ireland	42.5	−3.16	0.87	−0.66	−0.31	−0.04	+0.28	0.59	−0.63	−0.32	−0.04	+0.28	0.60
Lithuania	38.3	0.64	0.81	0.24	−0.31	−0.02	+0.33	0.64	0.27	−0.28	−0.04	+0.32	0.60
Sweden	32.0	−1.43	0.85	−1.78	−0.27	−0.03	+0.32	0.59	−0.74	−0.36	−0.17	+0.03	0.39
Luxembourg	27.1	−0.32	0.80	−0.25	−0.26	−0.03	+0.24	0.50	−0.22	−0.28	−0.04	+0.24	0.52
Denmark	26.5	−4.00	0.87	−1.41	−0.34	−0.05	+0.27	0.61	−1.09	0.00	0.00	0.00	0.00
Bulgaria	24.5	1.21	0.74	−2.07	−0.24	−0.03	+0.27	0.50	−0.88	−0.06	−0.01	+0.07	0.13
Estonia	21.4	0.00	0.76	−1.91	−0.24	−0.01	+0.26	0.50	−0.29	0.00	0.00	0.00	0.00
Mean (adjusting)	73.7	2.55	1.31	0.79	−0.36	+0.09	+0.65	1.01	1.23	−0.26	+0.10	+0.63	0.89

Source: Author's calculations. Note: Column (1) is debt in 2024. Column (2) is the adjustment that the baseline binding target requires, relative to the SPB in 2024. Column (3) is the band of the SPB required by the own criterion: the debt criterion of the baseline scenario. Base is the baseline target. P25, Median and P75 are the 25th, 50th and 75th percentiles of the change of the target across 500 joint draws, relative to the baseline. Band is the difference between P75 and P25. The last row gives the mean across the 21 countries whose binding target requires a positive adjustment.

the post-adjustment stochastic evaluation window from five to ten years; third, beginning the historical shock sample in 1990 or 2010 rather than 2000; fourth, estimating shocks at an annual rather than quarterly frequency; fifth, drawing shocks via a vector autoregression (VAR) bootstrap as in Darvas et al. (2025a) rather than from a multivariate normal distribution, which mirrors the Commission's planned refinements to incorporate shock persistence and fat tails (Bec et al., 2025); and sixth, winsorising the historical shock series at the 1st and 99th percentiles, or not at all, instead of at the 5th and 95th percentiles.

Although these econometric choices substantially alter the standalone SPB required by the stochastic criterion itself, almost none of the variation passes through to the overall DSA targets. Relaxing winsorisation to the 1st and 99th percentiles raises the standalone stochastic SPB by an average of 0.52 percentage points, removing winsorisation entirely raises it by 0.83 points, starting shocks during the adjustment period raises it by 0.20 points, and switching to annual shocks lowers it by 0.14 points. Because the stochastic criterion binds only for Finland in the baseline and lies well below the deterministic criteria in most other member states, however,

average DSA targets barely move. Relaxing winsorisation to the 1st and 99th percentiles raises the DSA target in France (0.36 points), Finland (0.27 points), and Austria (0.24 points), while removing winsorisation raises the targets of France and Austria by 0.63 points and those of Finland and Italy by 0.38 and 0.17 points. The VAR bootstrap raises Finland's DSA target by 0.46 points, while none of the remaining specifications alters any country's target by more than 0.11 points.

Similarly, introducing stochastic shocks during the adjustment period has a muted effect on its own, as proposed by Wieland (2024) and Erce (2025), because the criterion still compares debt five years after the adjustment period with debt at the end of the adjustment period, moving no country's DSA target by more than 0.1 points unless combined with a ten-year horizon (which raises Greece's target by 0.54 points) or a higher probability threshold, as shown next.

4.2 Calibration of risk tolerance

Unlike forecast distributions or econometric specifications, the parameters in the fourth group reflect normative choices regarding how much risk the fiscal framework is willing to accept: the probability threshold of the stochastic criterion, the shock magnitudes in the three deterministic stress tests, and the number of post-adjustment years over which projected changes in ageing costs are included in the SPB (see also Blanchard et al., 2021). Rather than assigning empirical distributions to these policy settings, Table 3b evaluates how the required SPB responds to plausible alternative calibrations, both individually (including a grid interacting the stochastic threshold, start year, and horizon) and through composite lenient and strict scenarios that bundle the lower and upper settings together.

Changing the stochastic probability threshold from 70 to 80 percent increases the standalone SPB required by the stochastic criterion by an average of 0.58 percentage points (and by more than 0.1 points in all 15 reference-trajectory countries with adjustment needs), while lowering the threshold to 60 percent reduces the standalone requirement by 0.49 points. Once again, however, the pass-through to the overall DSA target is small under the baseline projection window, with an 80 percent threshold raising the average DSA target by just 0.03 points and affecting only Finland and Austria by more than 0.1 points. Only when an 80 percent threshold is combined with starting the shocks in the first adjustment year does the average DSA target rise by 0.15 points, driven by increases of 1.48 points in Greece and 0.60 points in Finland (reaching 2.0 points in Greece when further combined with a ten-year horizon).

Among the deterministic stress tests, the calibration of the adverse $r - g$ test is far more influential because it binds for nine countries in the baseline. Increasing the simultaneous interest-rate and growth shocks from 0.5 to 0.75 percentage points raises the standalone stress-test SPB by 0.28 percentage points on average, lifting the overall DSA target by 0.16 points (and up to 0.49 points in Italy) and the binding target by 0.12 points, whereas reducing the shocks to 0.25 points lowers the average DSA target by 0.06 points. By contrast, increasing the lower-SPB

shock from 0.5 to 0.75 points raises the average DSA target by only 0.06 points, and varying the financial stress shock leaves all targets unchanged.

Although the stress-test shock sizes are policy conventions rather than empirical parameters, comparing them against the historical forecast error distributions from Section 3 reveals the implicit degree of risk protection they provide. The baseline adverse $r - g$ test increases the interest-growth differential by 1 percentage point in total. In the historical data, the difference between the ten-year forward-rate error and the five-year growth forecast error exceeded 1 percentage point in 30 percent of observations (both when combining the two error series independently and when pairing them by calendar year between 2000 and 2015). Subject to the caveat that the growth errors span five years while the rate errors span ten, the baseline $r - g$ stress test thus insures against roughly 70 percent of historical forecast errors in the interest-growth differential, while the 0.75-point shock (a 1.5-point shift in $r - g$) covers up to the 75th percentile. Strikingly, even though this 70 percent coverage of baseline forecast errors mirrors the 70 percent probability threshold of the stochastic criterion, the deterministic $r - g$ stress test requires a higher SPB than the stochastic criterion in nearly all member states.

Finally, the horizon over which governments must pre-fund projected changes in ageing costs (public spending on pensions, healthcare, and long-term care) shifts the targets for most countries. In the baseline, the SPB before ageing costs is held constant after the adjustment period, so any projected increase in ageing costs over the subsequent ten years reduces the headline primary balance one-for-one and must be offset upfront through a higher SPB target, whereas any projected decline in ageing costs lowers the target (Darvas et al., 2025b). This ten-year window assumes that governments will not adjust taxes or non-ageing expenditures after the adjustment period to absorb demographic pressures. Shortening this window from ten to five years reduces the average DSA target by 0.24 percentage points (and by up to 0.89 points in Spain), while raising the target in Bulgaria, Latvia, Estonia, Croatia, and Sweden, where ageing costs are projected to fall between the fifth and tenth post-adjustment years. Combining the five-year ageing window with the lower stress-test shocks and a 60 percent stochastic threshold in the lenient scenario lowers the average DSA target by 0.40 points, whereas the strict scenario raises it by 0.18 points (an asymmetry that reflects the fact that the ageing cost window cannot extend beyond the ten-year DSA horizon; see Table A3 in the Annex for country-level results).

Table 5 summarises the relative importance of macroeconomic uncertainty, stochastic modelling choices, and risk-tolerance calibration. On the lenient side, the 0.40 percentage point reduction in the average DSA target under the lenient risk-tolerance scenario matches the 0.36 point reduction under the 25th percentile of the joint forecast and parameter draws, driven largely by the shorter ageing-cost horizon. On the strict side, raising the adverse $r - g$ shock to 0.75 points increases the average DSA target by 0.16 points (roughly a quarter of the 0.65 point increase at the 75th percentile of the joint draws), while the composite strict scenario raises it by 0.18 points. By contrast, replacing normal shocks with the VAR bootstrap in the stochastic analysis moves the average DSA target by just 0.02 points. Taken together, the normative con-

Table 5: Uncertainty about the economy and tolerance for risk: mean change in the targets, four-year adjustment (percentage points of GDP)

Setting	DSA target	Binding target
Forecasts and model parameters		
Joint draws, 25th percentile (favourable)	−0.36	−0.26
Joint draws, 75th percentile (adverse)	+0.65	+0.63
Potential growth $T + 10$, −1 pp.	+0.42	+0.35
Real long-term rate $T + 10$, +1 pp.	+0.23	+0.17
Fiscal multiplier 1.0	+0.06	+0.22
Specification of the stochastic analysis		
VAR bootstrap	+0.02	0.00
No winsorisation	+0.09	+0.04
Start in first adjustment year	0.00	0.00
Risk tolerance		
Probability threshold 60%	−0.01	0.00
Probability threshold 80%	+0.03	+0.01
Threshold 80%, start in first adjustment year	+0.15	+0.09
Adverse $r - g$ shocks 0.25 pp.	−0.06	−0.05
Adverse $r - g$ shocks 0.75 pp.	+0.16	+0.12
Lower-SPB shock 0.25 pp.	0.00	−0.01
Lower-SPB shock 0.75 pp.	+0.06	+0.04
Ageing cost period 5 years	−0.24	−0.20
Lenient scenario	−0.40	−0.27
Strict scenario	+0.18	+0.13

Source: Author's calculations. Note: Mean change of the target relative to the baseline across the 21 countries with adjustment requirements. Divide by four for the change in the annual adjustment. Joint draws: 25th and 75th percentiles of the change across 500 draws of forecasts and model parameters. The other rows are individual checks and alternative settings from Tables 3a and 3b. Runs in which the debt sustainability safeguard cannot be met are left out of the binding target.

ventions governing stress-test sizes and ageing-cost horizons shape the fiscal targets by orders of magnitude similar to empirical forecast uncertainty, and far more than technical refinements to the stochastic fan chart.

5 Debt repayment profile and data vintage

Replacing the Commission's historical refinancing assumptions with actual residual maturity data significantly alters the fiscal targets for Greece and Finland, while leaving other member states largely unaffected. In the baseline model, the share of long-term debt maturing each year follows the Commission's inputs, which converge toward a historical average; in the alternative check, debt outstanding in 2024 is repaid according to Eurostat data on government debt by residual maturity (which are available for all member states except Hungary, Luxembourg, and the Netherlands). Because maturing debt is refinanced at prevailing market forward rates, which in both Greece and Finland exceed the average effective interest rate on the existing debt stock, the repayment schedule governs how quickly higher market rates pass through into total interest expenditure.

For Greece, the Commission's baseline inputs assume that the annual maturing share of long-term debt rises from roughly 4 percent in 2026 to between 9 and 11 percent after 2033, whereas Eurostat residual maturity data indicate that this share remains between 2 and 4 percent due to the long maturities of official stability mechanism loans. Because less debt needs to be rolled over at market rates under the Eurostat profile, Greece's projected average interest rate in 2038 is 3.0 percent rather than 3.5 percent for a given SPB path, leaving the 2038 debt ratio 3 percentage points of GDP lower and reducing Greece's DSA and binding targets by 0.57 and 0.42 percentage points, respectively. For Finland, the adjustment operates in the opposite direction: Eurostat data show roughly 14 percent of long-term debt maturing annually through 2028, compared with 3 to 4 percent in the Commission's baseline inputs. Faster refinancing at higher market rates raises Finland's projected 2038 average interest rate from 2.2 to 2.7 percent and its debt ratio by 3 percentage points of GDP, increasing its DSA target by 0.26 points and its binding target by 0.32 points, whereas no other country's target moves by more than 0.15 points.

To verify whether the sensitivity findings are specific to the initial 2024 prior guidance data, I re-estimate all individual forecast, parameter, and risk-tolerance checks using updated input data from September 2026, for which the four-year adjustment period spans 2027 to 2030. While the baseline target levels change across vintages (Table A5 in the Annex), the average sensitivity of the targets to each input remains remarkably stable: across all checks reported in Table A4, the correlation of mean target changes between the 2024 and 2026 vintages is 0.99 for the DSA target and 0.98 for the binding target, with mean effects differing by an average of only 0.02 percentage points (and by at most 0.06 points for higher potential growth and 0.13 points for the crisis multiplier).

These stable cross-country averages nevertheless mask notable shifts for individual member states as updated fiscal and macroeconomic conditions alter which constraint binds. Between the 2024 and 2026 vintages, the criterion setting the DSA target changes in four countries and the constraint setting the binding target changes in eight (Tables 1 and A5). In particular, the debt sustainability safeguard ceases to bind for Austria, lowering its binding SPB target from 2.22 to 1.16 percent of GDP, whereas Finland's binding target rises from 4.83 to 8.49 percent of GDP (requiring an annual SPB adjustment of 2.45 percentage points). Consequently, while the aggregate sensitivity of the EU framework to its inputs is robust over time, the specific inputs that matter for an individual member state shift whenever its binding constraint changes.

6 Conclusions

The sensitivity of a member state's fiscal adjustment target to the inputs of the Commission's DSA hinges on which regulatory criterion binds. Because macroeconomic forecasts and structural parameters shift the entire baseline debt trajectory, they affect all deterministic criteria simultaneously and move the targets for most countries. Drawing forecasts and model parameters jointly from historical forecast error distributions and published empirical ranges, I find

that the average DSA target across the 21 member states with positive adjustment needs falls by 0.36 percentage points of GDP at the 25th percentile and rises by 0.65 points at the 75th percentile, relative to an average baseline adjustment requirement of 2.6 points over four years. An adverse draw at the 75th percentile thus shifts the four-year target by roughly as much as the entire cumulative deviation permitted in the control account, even before accounting for the historical optimism bias of official growth projections (Frankel and Schreger, 2013), which I remove by demeaning the forecast error series.

Within the range of alternatives considered, normative choices governing the framework's risk tolerance move the targets by magnitudes comparable to empirical forecast uncertainty, particularly on the lenient side, where shortening the ageing-cost window and reducing stress-test shocks lowers the average DSA target by 0.40 percentage points, while raising the adverse $r - g$ shock from 0.5 to 0.75 points increases the average target by 0.16 points. By contrast, the econometric specification of the stochastic analysis has almost no effect on average targets because the stochastic criterion binds solely for Finland in the baseline, meaning that the Commission's planned technical refinements to the stochastic fan chart will leave most targets unchanged (Bec et al., 2025). In practice, the framework insures against baseline projection errors primarily through the deterministic stress tests: the baseline adverse $r - g$ shock of 0.5 percentage points covers roughly 70 percent of historical forecast errors in the interest-growth differential, matching the 70 percent probability threshold of the stochastic criterion while imposing a stricter SPB requirement in almost all countries.

These findings point to three priorities for the upcoming review of the DSA methodology. First, the calibration of the deterministic stress tests and the horizon for pre-funding ageing costs warrant at least as much scrutiny as the stochastic simulations, which are theoretically appealing but empirically non-binding for 26 of the 27 member states. Second, because the debt sustainability safeguard evaluates debt reduction strictly during the adjustment years, it interacts adversely with the fiscal multiplier and can require self-defeating consolidations when multipliers are high; evaluating debt reduction over a post-adjustment window or removing the safeguard would eliminate this feedback loop. Third, the degree of risk that the rules implicitly tolerate through stress-test shock sizes, probability thresholds, and ageing-cost windows should be justified explicitly as a policy choice.

Finally, future methodological reviews should consider how endogenous growth effects alter the fiscal arithmetic. Potential output in the DSA follows the EU's commonly agreed methodology and is invariant to the composition of fiscal adjustment. In Gottschalk et al. (2025), we show that allowing productive public investment in national medium-term plans to raise potential growth can ease adjustment requirements, with Germany's infrastructure package reducing its SPB target from 0.99 to 0.84 percent of GDP. Conversely, the current DSA also omits hysteresis channels through which sharp fiscal consolidations can depress potential output over the medium run (Fatás and Summers, 2018; Gechert et al., 2019).

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Annex

Table A1: Distribution of past forecast errors

Assumption	Source of errors	Sample	*N*	Mean error	P10	P25	P75	P90
Real short-term rate, $T+10$	Bundesbank: 1-year rate in ten years minus forward rate	Germany, year ends 1986-2015	30	−4.13	−1.61	−1.22	+0.40	+3.11
Real long-term rate, $T+10$	Bundesbank: 10-year rate in ten years minus 10Y10Y forward rate	Germany, year ends 1986-2015	30	−2.98	−1.53	−1.24	+0.89	+2.36
Inflation, $T+10$	IMF WEO: CPI inflation, average error over five years	EU27, autumn vintages 2000-2018	513	+0.14	−1.33	−0.74	+0.64	+1.35
Potential growth, $T+10$	IMF WEO: real GDP growth, average error over five years	EU27, autumn vintages 2000-2018	513	−0.70	−2.53	−0.84	+1.08	+2.31
Change in ageing costs, 2024–2038	Ageing Report 2024: scenarios except pension policy, change in ageing costs 2024-2038	EU27, 8 scenarios	216	–	−0.38	−0.24	+0.14	+0.62

Source: Author's calculations based on Ech-charfi (2024), the Deutsche Bundesbank term structure of interest rates and European Commission (2024a). Note: Errors are realised minus forecast values, in percentage points. WEO errors are averages over the five forecast years of each autumn vintage, demeaned by country. Forward-rate errors compare the forward rate at each year end with the rate realised ten years later, demeaned. Ageing costs: deviation of each scenario from the baseline change in the total cost of ageing between 2024 and 2038, in percentage points of GDP, for all countries and all scenarios except the pension policy scenarios. Mean error: average error before demeaning. P10 to P90: percentiles of the demeaned errors. The individual checks shift each forecast by a round amount close to half the distance between the 25th and the 75th percentile; the joint draws use the full distribution.

Table A2a: Change in the SPB relative to the baseline: forecasts and model parameters, seven-year adjustment (percentage points of GDP)

		Own criterion		DSA target		Binding target	
Input	Value	Mean	Largest	Mean	Largest	Mean	Largest
1. Forecasts							
Real short-term rate, $T + 10$	−1 pp.	−0.04	−0.15 ITA	−0.04	−0.15 ITA	−0.03	−0.18 ITA
	+1 pp.	+0.04	+0.17 ITA	+0.04	+0.15 ITA	+0.04	+0.15 ITA
Real long-term rate, $T + 10$	−1 pp.	−0.30	−0.65 GRC	−0.27	−0.47 ITA	−0.17	−0.42 ESP
	+1 pp.	+0.30	+0.60 GRC	+0.30	+0.57 ITA	+0.21	+0.48 GRC
Inflation, $T + 10$	−0.75 pp.	+0.15	+0.50 ITA	+0.10	+0.47 GRC	+0.06	+0.42 GRC
	+0.75 pp.	−0.13	−0.41 ITA	+0.01	+0.30 ITA	+0.01	−0.28 FIN
Potential growth, $T + 10$	−1 pp.	+0.54	+1.34 ITA	+0.47	+1.10 ITA	+0.36	+0.86 ITA
	+1 pp.	−0.51	−1.16 ITA	−0.29	−0.60 FIN	−0.16	−0.49 FIN
Ageing costs	−0.2 pp.	−0.07	−0.15 AUT	−0.08	−0.14 AUT	−0.05	−0.14 CZE
	+0.2 pp.	+0.08	+0.15 AUT	+0.09	+0.14 AUT	+0.06	+0.14 CZE
Real short-term rate, $T + 30$	−0.5 pp.	0.00	−0.02 PRT	0.00	−0.02 PRT	0.00	0.00
	+0.5 pp.	0.00	+0.02 PRT	0.00	+0.02 PRT	0.00	0.00
Real long-term rate, $T + 30$	−0.5 pp.	−0.02	−0.07 GRC	−0.02	−0.05 AUT	0.00	−0.04 ITA
	+0.5 pp.	+0.02	+0.07 GRC	+0.02	+0.06 GRC	+0.01	+0.07 ESP
Inflation, $T + 30$	−0.25 pp.	+0.02	+0.06 GRC	+0.01	+0.05 GRC	+0.01	+0.07 PRT
	+0.25 pp.	−0.02	−0.06 GRC	−0.01	−0.05 BEL	0.00	−0.02 BEL
2. Model parameters							
Fiscal multiplier	0.5	−0.01	−0.07 SVK	−0.02	−0.06 SVK	−0.07	−0.42 FIN
	1	+0.01	+0.07 SVK	+0.03	+0.23 FRA	+0.08	+0.56 FIN
	1.5	+0.04	+0.21 SVK	+0.13	+0.79 FRA	+0.31	+2.10 FIN
Multiplier persistence	2 years	0.00	+0.13 ITA	−0.01	−0.05 FIN	−0.05	−0.28 FIN
	4 years	+0.02	+0.05 SVK	+0.03	+0.16 FRA	+0.05	+0.35 FIN
	5 years	+0.03	+0.11 FRA	+0.05	+0.33 FRA	+0.10	+0.63 FIN
Budget semi-elasticity	×0.9	−0.01	−0.02 ROU	−0.01	−0.04 FRA	−0.03	−0.14 FIN
	×1.1	+0.01	+0.02 ROU	+0.01	+0.06 FRA	+0.02	+0.21 FIN

Source: Author's calculations. Note: See Table 3a. With a seven-year adjustment, the debt sustainability safeguard can be met in all runs.

Table A2b: Change in the SPB relative to the baseline: estimation of risk, risk tolerance and data, seven-year adjustment (percentage points of GDP)

		Own criterion		DSA target		Binding target	
Input	Value	Mean	Largest	Mean	Largest	Mean	Largest
3. Estimation of risk							
Start of stochastic projection	first adjustment year	+0.48	+1.98 GRC	+0.14	+1.96 GRC	+0.10	+1.82 GRC
Horizon of stochastic projection	10 years	0.00	−0.80 FIN	0.00	+0.09 GRC	0.00	+0.06 GRC
Start of shock sample	1990	−0.02	−0.08 GRC	0.00	0.00	0.00	0.00
	2010	+0.01	+0.34 SVN	0.00	0.00	0.00	0.00
Shock frequency	annual	−0.14	−0.56 MLT	0.00	0.00	0.00	0.00
Shock distribution	VAR bootstrap	+0.06	−0.70 ROU	+0.06	+0.60 GRC	+0.03	+0.54 GRC
Winsorised shocks	P1 and P99	+0.52	+1.08 ESP	+0.10	+0.53 FRA	+0.04	+0.25 ITA
	off	+0.82	+1.64 ESP	+0.21	+0.99 GRC	+0.11	+0.84 GRC
4. Risk tolerance							
Threshold, start, horizon	60%, after adj., 5 years	−0.50	−0.90 GRC	0.00	0.00	0.00	0.00
	80%, after adj., 5 years	+0.58	+1.03 GRC	+0.11	+1.00 GRC	+0.06	+0.84 GRC
	60%, after adj., 10 years	−0.40	−1.09 FIN	0.00	0.00	0.00	0.00
	80%, after adj., 10 years	+0.45	+0.98 GRC	+0.07	+0.96 GRC	+0.05	+0.84 GRC
	60%, first adj. year, 5 years	−0.30	−0.71 MLT	0.00	0.00	0.00	0.00
	80%, first adj. year, 5 years	+1.37	+4.26 GRC	+0.42	+4.05 GRC	+0.30	+3.92 GRC
	60%, first adj. year, 10 years	−0.24	−1.04 FIN	0.00	0.00	0.00	0.00
	70%, first adj. year, 10 years	+0.33	+1.45 GRC	+0.07	+1.42 GRC	+0.06	+1.26 GRC
	80%, first adj. year, 10 years	+0.99	+3.18 GRC	+0.26	+3.16 GRC	+0.20	+3.01 GRC
Adverse $r - g$ shock	0.25 pp.	−0.27	−0.49 ITA	−0.04	−0.21 BEL	−0.01	−0.14 ESP
	0.75 pp.	+0.26	+0.48 ITA	+0.16	+0.44 GRC	+0.09	+0.36 GRC
Lower-SPB shock	0.25 pp.	−0.20	−0.28 ITA	0.00	−0.06 DEU	0.00	0.00
	0.75 pp.	+0.21	+0.28 ITA	+0.06	+0.23 PRT	+0.04	+0.21 PRT
Financial stress shock	0.5 pp.	−0.03	−0.13 ITA	0.00	0.00	0.00	0.00
	1.5 pp.	+0.03	+0.13 ITA	0.00	0.00	0.00	0.00
Ageing cost period	5 years	−0.17	−0.87 ESP	−0.19	−0.85 CZE	−0.12	−0.77 ESP
All settings	lenient	−0.17	−0.87 ESP	−0.36	−1.15 ESP	−0.16	−1.12 ESP
	strict	0.00	0.00	+0.21	+1.00 GRC	+0.12	+0.84 GRC
Data							
Repayment profile	Eurostat profile	−0.01	−0.48 GRC	−0.01	−0.36 GRC	+0.02	+0.42 FIN

Source: Author's calculations. Note: See Table 3b. With a seven-year adjustment, the debt sustainability safeguard can be met in all runs.

Table A3: Effect of selected settings of risk tolerance by country, DSA target, four-year adjustment (percentage points of GDP)

	Thr. 80% (1)	Early start (2)	Early, 80% (3)	$r - g$ 0.75 pp. (4)	Low SPB 0.75 pp. (5)	Ageing 5 yrs (6)	Lenient (7)	Strict (8)
Greece	+0.02	0.00	+1.48	+0.44	0.00	−0.41	−0.62	+0.44
Italy	0.00	0.00	+0.14	+0.49	0.00	0.00	−0.38	+0.49
France	+0.04	0.00	+0.35	+0.40	+0.02	0.00	−0.08	+0.40
Spain	0.00	0.00	0.00	+0.33	+0.05	−0.89	−1.25	+0.33
Belgium	0.00	0.00	0.00	+0.36	0.00	−0.49	−0.70	+0.36
Portugal	+0.01	0.00	+0.23	+0.27	+0.19	−0.77	−1.06	+0.27
Finland	+0.46	+0.09	+0.60	+0.10	+0.18	0.00	−0.38	+0.46
Austria	+0.11	0.00	+0.32	+0.31	+0.11	−0.22	−0.55	+0.31
Hungary	+0.01	0.00	+0.01	+0.19	+0.18	−0.50	−1.07	+0.19
Cyprus	0.00	0.00	0.00	+0.13	+0.17	−0.23	−0.42	+0.17
Slovenia	0.00	0.00	0.00	+0.12	+0.19	−0.20	−0.39	+0.19
Germany	0.00	0.00	0.00	+0.19	+0.17	−0.06	−0.26	+0.19
Croatia	0.00	0.00	0.00	0.00	0.00	+0.05	+0.05	0.00
Slovakia	0.00	0.00	0.00	+0.11	+0.19	−0.15	−0.34	+0.19
Poland	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Malta	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Romania	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Netherlands	0.00	0.00	0.00	0.00	0.00	−0.65	−0.65	0.00
Czechia	0.00	0.00	0.00	0.00	0.00	−0.66	−0.66	0.00
Latvia	0.00	0.00	0.00	0.00	0.00	+0.20	+0.20	0.00
Ireland	0.00	0.00	0.00	0.00	0.00	−0.46	−0.46	0.00
Lithuania	0.00	0.00	0.00	0.00	0.00	−0.54	−0.54	0.00
Sweden	0.00	0.00	0.00	0.00	0.00	+0.05	+0.05	0.00
Luxembourg	0.00	0.00	0.00	0.00	0.00	−0.70	−0.70	0.00
Denmark	0.00	0.00	0.00	0.00	0.00	−0.16	−0.16	0.00
Bulgaria	0.00	0.00	0.00	0.00	0.00	+0.26	+0.26	0.00
Estonia	0.00	0.00	0.00	0.00	0.00	+0.09	+0.09	0.00
Mean	+0.03	0.00	+0.15	+0.16	+0.06	−0.24	−0.40	+0.18

Source: Author's calculations. Note: Change in the DSA target relative to the baseline. (1) Probability threshold of 80 percent. (2) Stochastic projection starting in the first adjustment year. (3) Both. (4) Adverse $r - g$ shocks of 0.75 percentage points. (5) Lower-SPB shock of 0.75 percentage points. (6) Ageing cost period of five years. (7) and (8) Lenient and strict scenarios, see Table 3b. Mean: mean across the 21 countries with adjustment requirements. Countries are ordered by debt in 2024.

Table A4: Mean change in the SPB by individual check with the 2024 and the September 2026 data, four-year adjustment (percentage points of GDP)

		Own criterion		DSA target		Binding target	
Setting	Alternative	2024	2026	2024	2026	2024	2026
1. Forecasts							
Real short-term rate, $T + 10$	−1 pp.	−0.04	−0.04	−0.04	−0.04	−0.03	−0.03
	+1 pp.	+0.04	+0.04	+0.04	+0.04	+0.03	+0.04
Real long-term rate, $T + 10$	−1 pp.	−0.26	−0.26	−0.23	−0.26	−0.17	−0.20
	+1 pp.	+0.25	+0.26	+0.23	+0.25	+0.17	+0.19
Inflation, $T + 10$	−0.75 pp.	+0.18	+0.20	+0.11	+0.14	+0.10	+0.12
	+0.75 pp.	−0.16	−0.16	−0.04	−0.07	−0.03	−0.09
Potential growth, $T + 10$	−1 pp.	+0.54	+0.56	+0.42	+0.48	+0.35	+0.39
	+1 pp.	−0.51	−0.54	−0.29	−0.35	−0.20	−0.29
Ageing costs	−0.2 pp.	−0.09	−0.09	−0.08	−0.09	−0.06	−0.07
	+0.2 pp.	+0.09	+0.09	+0.08	+0.10	+0.06	+0.04
Real short-term rate, $T + 30$	−0.5 pp.	0.00	0.00	0.00	0.00	0.00	0.00
	+0.5 pp.	0.00	0.00	0.00	0.00	0.00	0.00
Real long-term rate, $T + 30$	−0.5 pp.	−0.01	−0.01	−0.01	−0.01	−0.01	0.00
	+0.5 pp.	+0.01	+0.01	+0.01	+0.01	0.00	+0.01
Inflation, $T + 30$	−0.25 pp.	+0.01	+0.01	+0.01	+0.01	0.00	+0.01
	+0.25 pp.	−0.01	−0.01	−0.01	−0.01	−0.01	−0.01
2. Model parameters							
Fiscal multiplier	0.5	−0.01	−0.01	−0.04	−0.02	−0.13	−0.16
	1	+0.01	+0.01	+0.06	+0.03	+0.22	+0.26
	1.5	+0.04	+0.04	+0.28	+0.24	+0.95	+0.82
Multiplier persistence	2 years	−0.01	−0.01	−0.03	−0.02	−0.08	−0.10
	4 years	+0.01	+0.01	+0.04	+0.02	+0.10	+0.12
	5 years	+0.03	+0.03	+0.08	+0.06	+0.18	+0.23
Budget semi-elasticity	×0.9	−0.01	−0.01	−0.01	−0.01	−0.05	−0.05
	×1.1	+0.01	+0.01	+0.01	+0.01	+0.06	+0.06
3. Estimation of risk							
Start of stochastic projection	first adjustment year	+0.20	+0.23	0.00	+0.01	0.00	0.00
Horizon of stochastic projection	10 years	+0.14	+0.14	−0.01	−0.01	0.00	0.00
Start of shock sample	1990	−0.02	−0.01	0.00	0.00	0.00	0.00
	2010	+0.01	+0.01	0.00	0.00	0.00	0.00
Shock frequency	annual	−0.14	−0.17	0.00	0.00	0.00	0.00
Shock distribution	VAR bootstrap	+0.06	+0.02	+0.02	+0.03	0.00	0.00
Winsorised shocks	P1 and P99	+0.52	+0.51	+0.04	+0.02	+0.02	+0.01
	off	+0.83	+0.81	+0.09	+0.11	+0.04	+0.06
4. Risk tolerance							
Probability threshold	60%	−0.49	−0.54	−0.01	−0.01	0.00	0.00
	80%	+0.58	+0.62	+0.03	+0.07	+0.01	+0.02
Adverse $r - g$ shock	0.25 pp.	−0.28	−0.27	−0.06	−0.06	−0.05	−0.05
	0.75 pp.	+0.28	+0.27	+0.16	+0.18	+0.12	+0.14
Financial stress shock	0.5 pp.	−0.02	−0.02	0.00	0.00	0.00	0.00
	1.5 pp.	+0.02	+0.02	0.00	0.00	0.00	0.00
Lower-SPB shock	0.25 pp.	−0.23	−0.23	0.00	−0.01	−0.01	−0.01
	0.75 pp.	+0.23	+0.23	+0.06	+0.10	+0.04	+0.07
Ageing cost period	5 years	−0.22	−0.20	−0.24	−0.23	−0.20	−0.18
Scenario	lenient	−0.22	−0.20	−0.40	−0.43	−0.27	−0.32
	strict	0.00	0.00	+0.18	+0.23	+0.13	+0.16
Data							
Repayment profile	Eurostat profile	−0.01	+0.04	−0.01	+0.04	0.00	+0.04

Source: Author's calculations. Note: Mean change across the 21 countries with adjustment requirements in the 2024 data, relative to the baseline of each data vintage. 2024: Commission prior guidance data, adjustment from 2025 to 2028. 2026: September 2026 data, adjustment from 2027 to 2030.

Table A5: SPB targets with the September 2026 data (percent of GDP)

	2026		Four-year adjustment, 2027–2030					Change vs. 2024 data			Seven-year
	Debt	SPB	DSA target	DSA criterion	Binding target	Binding constraint	Annual adj.	DSA target	Binding target	Annual adj.	Binding target
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Greece	140.7	2.54	2.05	Adverse $r-g$ test: debt declines	2.54	Adjustment floor	0.00	−0.26	−0.04	−0.21	2.54
Italy	138.5	0.97	3.18	Adverse $r-g$ test: debt declines	3.21	DSA criteria	0.56	−0.11	−0.10	−0.53	3.00
France	118.1	−1.97	1.77	Adverse $r-g$ test: debt declines	1.79	DSA criteria	0.94	+1.03	+1.04	+0.01	2.49
Belgium	110.5	−1.96	1.87	Adverse $r-g$ test: debt declines	1.88	DSA criteria	0.96	+0.70	+0.67	+0.15	2.32
Spain	99.6	−0.40	2.82	Adverse $r-g$ test: debt declines	2.84	DSA criteria	0.81	+0.25	+0.25	−0.05	2.96
Finland	91.2	−1.31	1.83	Stochastic	8.49	Debt safeguard	2.45	+0.04	+3.66	+1.11	10.87
Portugal	87.6	2.20	2.59	Adverse $r-g$ test: debt declines	2.60	DSA criteria	0.10	+0.10	+0.08	+0.02	2.34
Austria	83.4	−1.72	1.13	Adverse $r-g$ test: debt declines	1.16	DSA criteria	0.72	+0.32	−1.06	−0.17	1.66
Hungary	75.1	−2.01	3.21	Adverse $r-g$ test: debt declines	3.23	DSA criteria	1.31	+0.57	+0.56	+0.65	3.66
Germany	65.8	−1.77	1.10	Adverse $r-g$ test: debt declines	2.75	Debt safeguard	1.13	+0.11	+0.29	+0.26	2.29
Slovenia	64.9	−2.40	1.00	Lower-SPB test: debt below 60% at end	1.04	DSA criteria	0.86	+0.52	+0.52	+0.42	1.17
Poland	64.5	−3.77	1.75	Adverse $r-g$ test: debt declines	1.79	DSA criteria	1.39	+1.18	+1.21	+0.59	2.32
Slovakia	63.7	−2.63	1.67	Lower-SPB test: debt below 60% at end	1.69	DSA criteria	1.08	+0.86	+0.87	−0.19	1.85
Romania	61.6	−2.39	1.46	Lower-SPB test: debt below 60% at end	1.49	DSA criteria	0.97	+1.20	+1.22	−0.20	2.23
Croatia	55.9	−1.74	−0.95	Baseline: debt below 60% at end	−0.14	Deficit resilience	0.40	−0.49	+0.22	0.00	0.01
Cyprus	50.4	2.52	0.10	Deficit below 3%	0.12	DSA criteria	−0.60	−0.25	−3.36	−0.60	−0.35
Latvia	48.8	−2.08	0.01	Lower-SPB test: debt below 60% at end	0.04	DSA criteria	0.53	+1.29	+0.18	+0.22	0.30
Netherlands	46.9	−0.45	−0.01	Deficit below 3%	−0.01	DSA criteria	0.11	−0.06	−0.08	−0.07	−0.31
Malta	46.2	−0.29	−0.94	Deficit below 3%	0.19	Deficit resilience	0.12	+0.29	+1.06	−0.39	0.06
Czechia	45.8	−1.33	0.98	Deficit below 3%	0.99	DSA criteria	0.58	+0.61	+0.60	+0.46	1.40
Lithuania	44.6	−1.05	0.47	Deficit below 3%	0.54	Deficit resilience	0.40	+0.23	+0.27	+0.24	0.55
Sweden	36.6	−1.35	−1.55	Deficit below 3%	−0.65	Deficit resilience	0.17	+0.23	+0.09	+0.53	−0.47
Ireland	32.4	2.09	−0.46	Deficit below 3%	−0.43	DSA criteria	−0.63	+0.20	+0.20	+0.16	−0.36
Bulgaria	32.3	−2.93	−1.18	Deficit below 3%	−1.17	DSA criteria	0.44	+0.89	−0.29	+0.14	−0.76
Luxembourg	29.2	0.75	0.18	Deficit below 3%	0.19	DSA criteria	−0.14	+0.42	+0.41	−0.06	0.05
Denmark	27.0	2.75	−1.29	Deficit below 3%	−1.25	Adjustment floor	−1.00	+0.12	−0.17	0.00	−0.84
Estonia	26.9	−3.20	−1.28	Deficit below 3%	−1.28	DSA criteria	0.48	+0.63	−0.99	+0.48	−0.89

Source: Author's calculations based on the September 2026 input data. Note: Columns (1) and (2) are debt and the SPB in 2026, the reference year of this data vintage. Columns (3) to (7) refer to a four-year adjustment period ending in 2030: the DSA target and the DSA criterion that sets it, the binding target and the constraint that sets it, and the annual adjustment required by the binding target. Columns (8) to (10) are the changes in the targets and in the annual adjustment relative to the baseline with the 2024 data, whose targets refer to 2028. Column (11) is the binding target at the end of a seven-year adjustment period (2033). Countries are ordered by debt in 2026.

Table A6: Fiscal multiplier checks with the persistent multiplier and the Commission's output gap rule, binding target, four-year adjustment (percent of GDP and percentage points)

	Austria		Finland		Germany	
	Persistent	Commission	Persistent	Commission	Persistent	Commission
Baseline target	2.22	2.58	4.83	5.99	2.46	2.78
Fiscal multiplier: 0.5	−0.64	−0.80	−1.00	−1.44	−0.52	−0.64
Fiscal multiplier: 1	+1.04	+1.52	+1.64	+2.60	+0.76	+1.00
Fiscal multiplier: 1.5	+8.56	inf.	inf.	inf.	+4.08	+7.00
Multiplier persistence: 2 years	−0.40	−0.64	−0.64	−1.36	−0.32	−0.56
Multiplier persistence: 4 years	+0.44	+0.48	+0.68	+1.04	+0.32	+0.40
Multiplier persistence: 5 years	+0.76	+0.88	+1.20	+1.88	+0.56	+0.68

Source: Author's calculations. Note: The first row is the binding target of the baseline. The other rows are the change in the binding target relative to the baseline. inf.: the debt sustainability safeguard cannot be met with an annual adjustment of up to 5 percentage points. Persistent: multiplier effect on the output gap fading over the persistence period, as in the rest of this paper. Commission: output gap closure rule of the Commission's prior guidance calculation sheets. The debt sustainability safeguard sets the binding target of the three countries in the baseline.

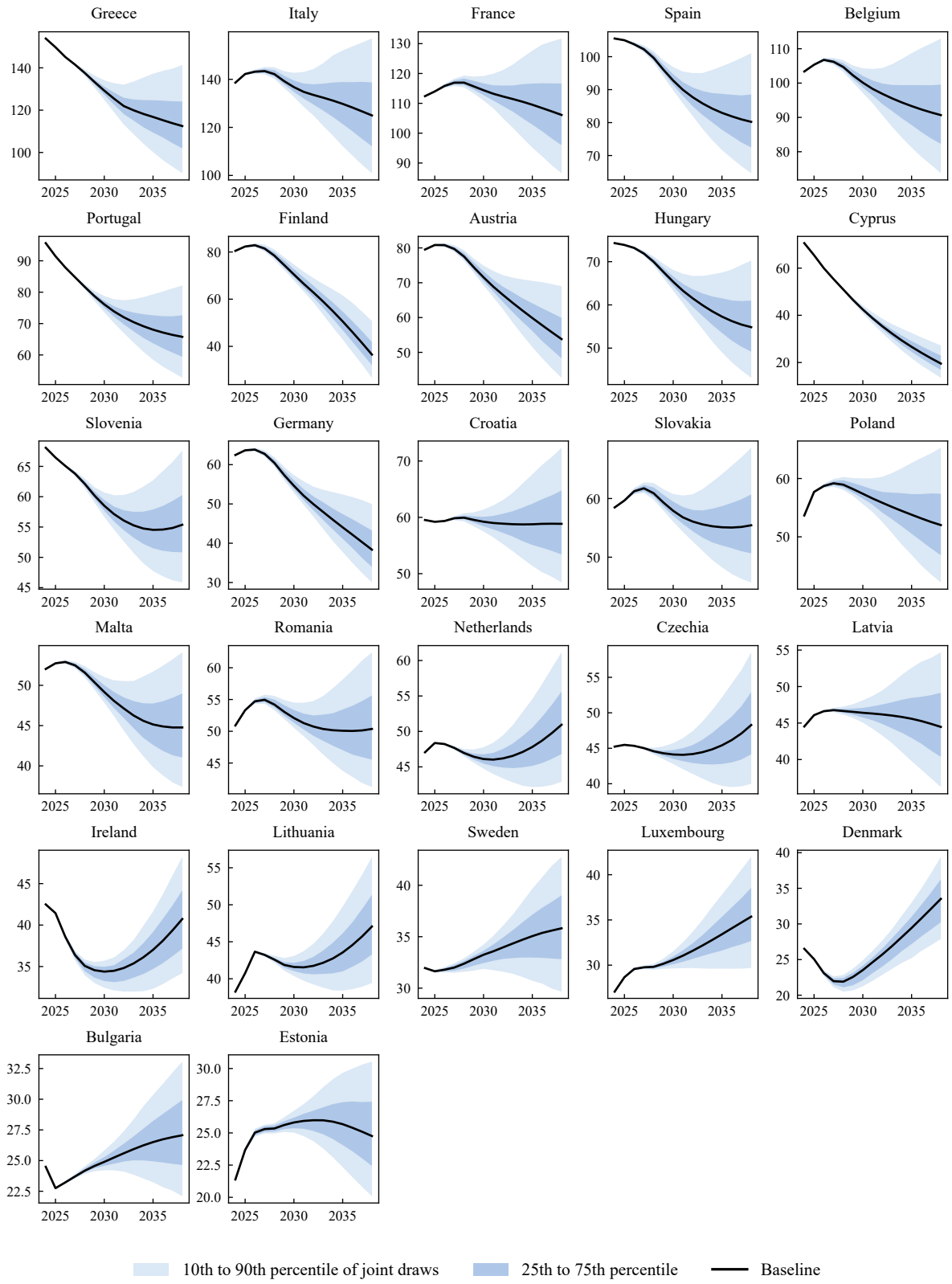


Figure A1: Debt under the baseline SPB path in the joint draws, four-year adjustment (percent of GDP)

Source: Author's calculations. Note: Debt when the SPB follows the path of the baseline binding target in all draws, with the adjustment from 2025 to 2028 including the minimum adjustment of the EDP and the deficit resilience safeguard. Only the forecasts and the model parameters differ across draws. The black line shows the baseline. The shaded areas show the range between the 10th and the 90th percentile (light) and between the 25th and the 75th percentile (dark) of debt in each year across the 500 joint draws of forecasts and model parameters. Countries are ordered by debt in 2024.